

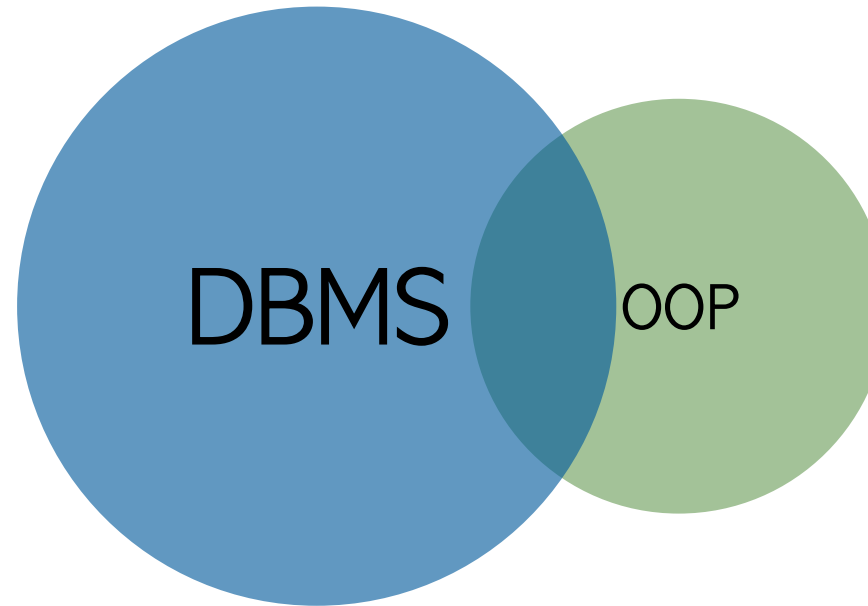
1

Outline: Week 02 A: Monday

- I) Background Requirement
- II) Topics
- III) Assessments & Team Project
- III) A Brightspace Tour

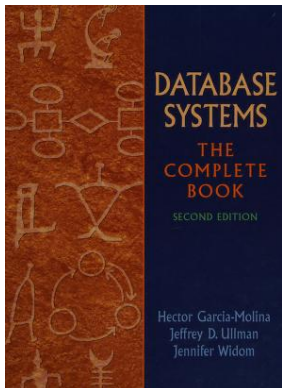
Background

ER Diagram
Relational Algebra
SQL
DBMS e.g., MySQL

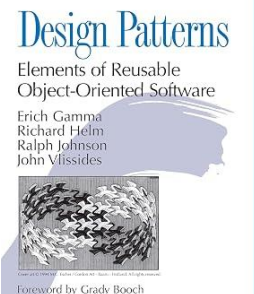


Object-Oriented Programming
Software Layered Architecture

Data Access Layer
Relational to Object Mapping
User Interface



Database Systems: The Complete Book
by Hector Garcia-Molina, Jeffrey Ullman, Jennifer Widom
2008

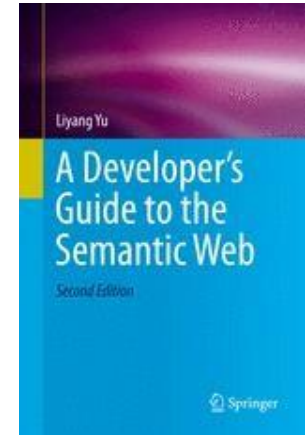
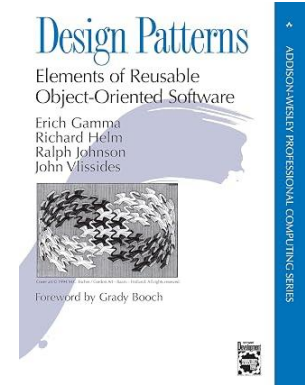
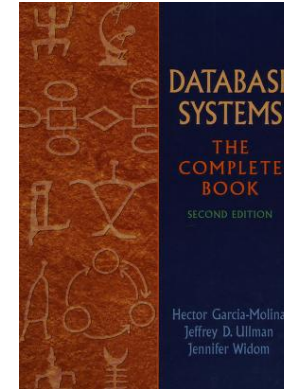


Design Patterns: Elements of Reusable Object-Oriented Software
by Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides
1994 (Gang of 4 Design Patterns)

Topics

- Conceptual Modeling by Entity-Relationship
- Conceptual Modeling by Classes in OOP
- Logical Modeling by Objects in OOP
- Logical Modeling by Relational
- Physical Modeling by SQL
- Logical Modeling by Resource Description Framework (RDF)
- Conceptual Modeling by RDF Schema (RFDS)
- Conceptual Modeling by Web Ontology Language (OWL)
- Logical Modeling by OWL
- Physical Modeling by SPARQL

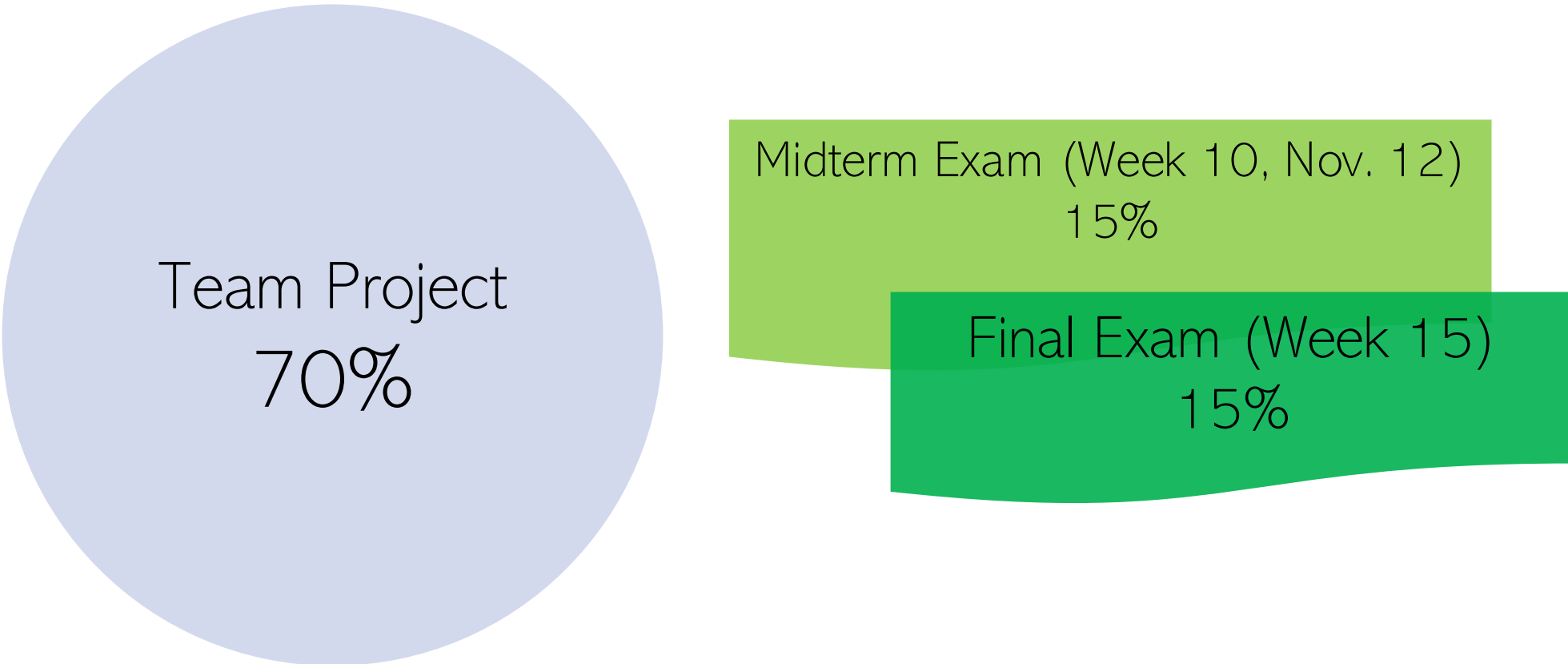
“sparkle”, a recursive acronym for SPARQL Protocol and RDF Query Language



A Developer's Guide to the Semantic Web,
by Liyang Yu, 2014

4

Outline: W02A



Team Project
70%

Midterm Exam (Week 10, Nov. 12)
15%

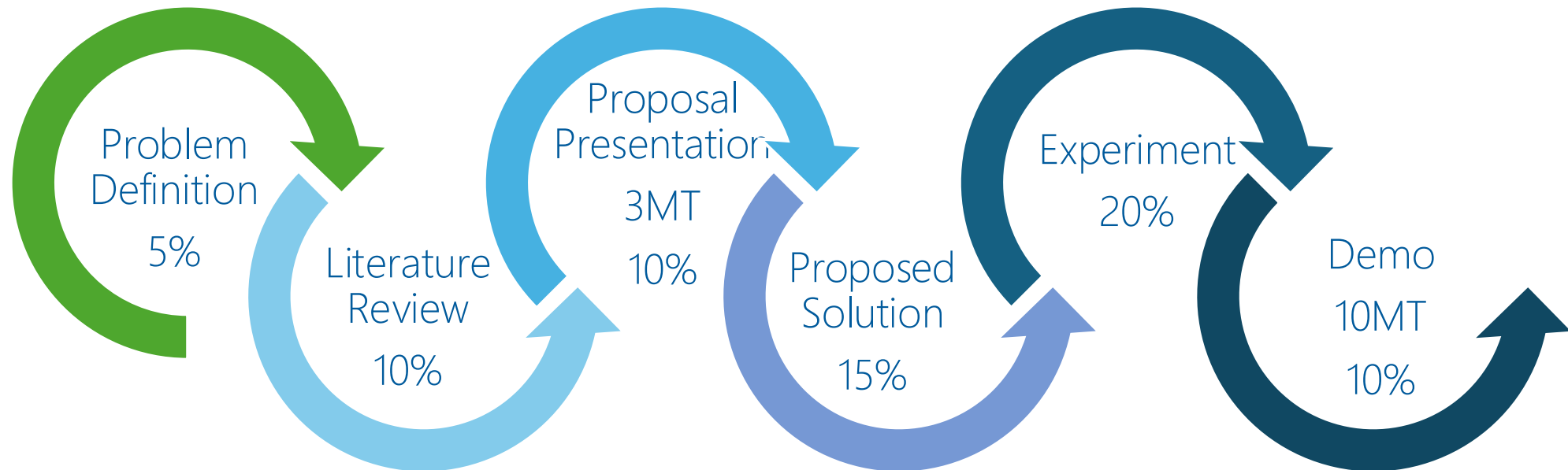
Final Exam (Week 15)
15%

5

Team Project

Min. 2, Max. 3 Members

Course-long Project vs. Final Project

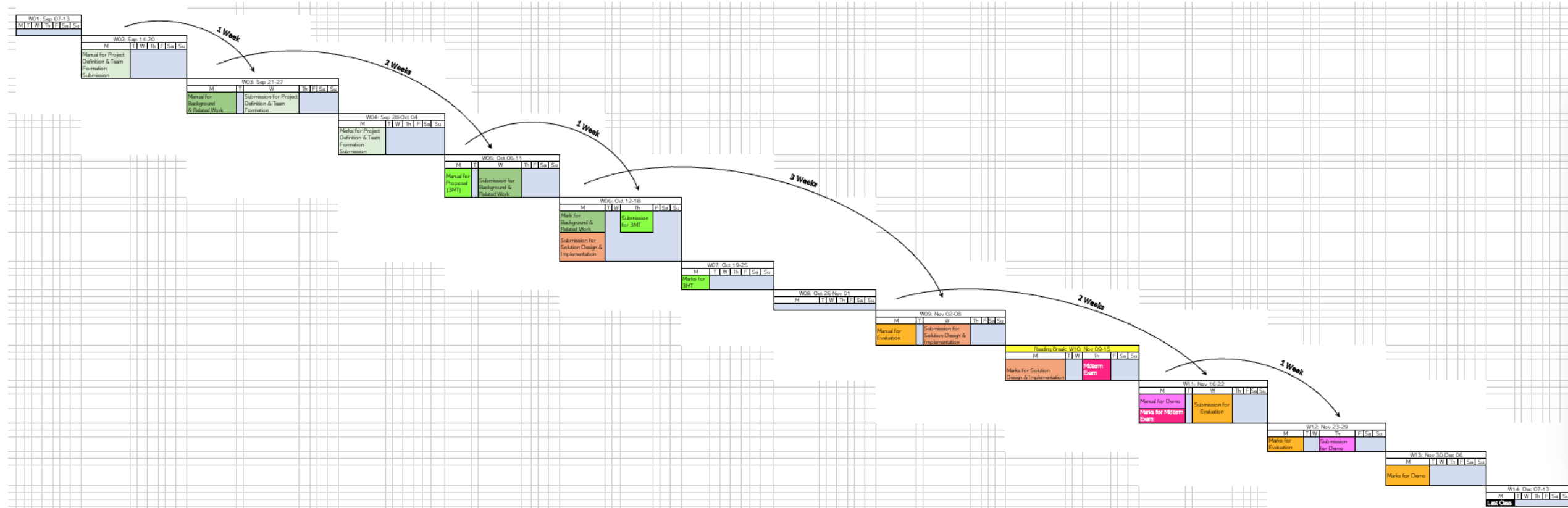


6

Weekly Schedule

MON	TUE	WED	THU	FRI	SAT	SUN	MON
New Submission Manual Prev. Submission Marks Release		Submission Deadline Midnight					
Lecture A: 8:30-9:50 Elliot Building 061: Hossein			Lecture B: 8:30-9:50 Elliot Building 061: Hossein				
Office A: 10:00-11:00 ECS613: Hossein			Office B: 10:00-11:00 ECS613: Hossein				
		Marking					Marking - 12:00 AM
	Monitor Discussion Board						
Link Lecture Video			Link Lecture Video				Release Marks, Stats, and Make Announcer

Timeline



Team Project

Take a Small but Novel Step!

Data Modeling Security and Safety Events

Data Modeling Security and Safety Events in Victoria

Data Modeling Security and Safety Events in University of Victoria

Data Modeling Security and Safety Events in University of Victoria

Team Project

Example: An Ontology for Describing Security Events

https://hosseinfani.github.io/res/papers/2015_SEKE_An_Ontology_for_Describing_Security_Events.pdf

<https://www.slideshare.net/slideshow/an-ontology-for-describing-security-events/238327365>

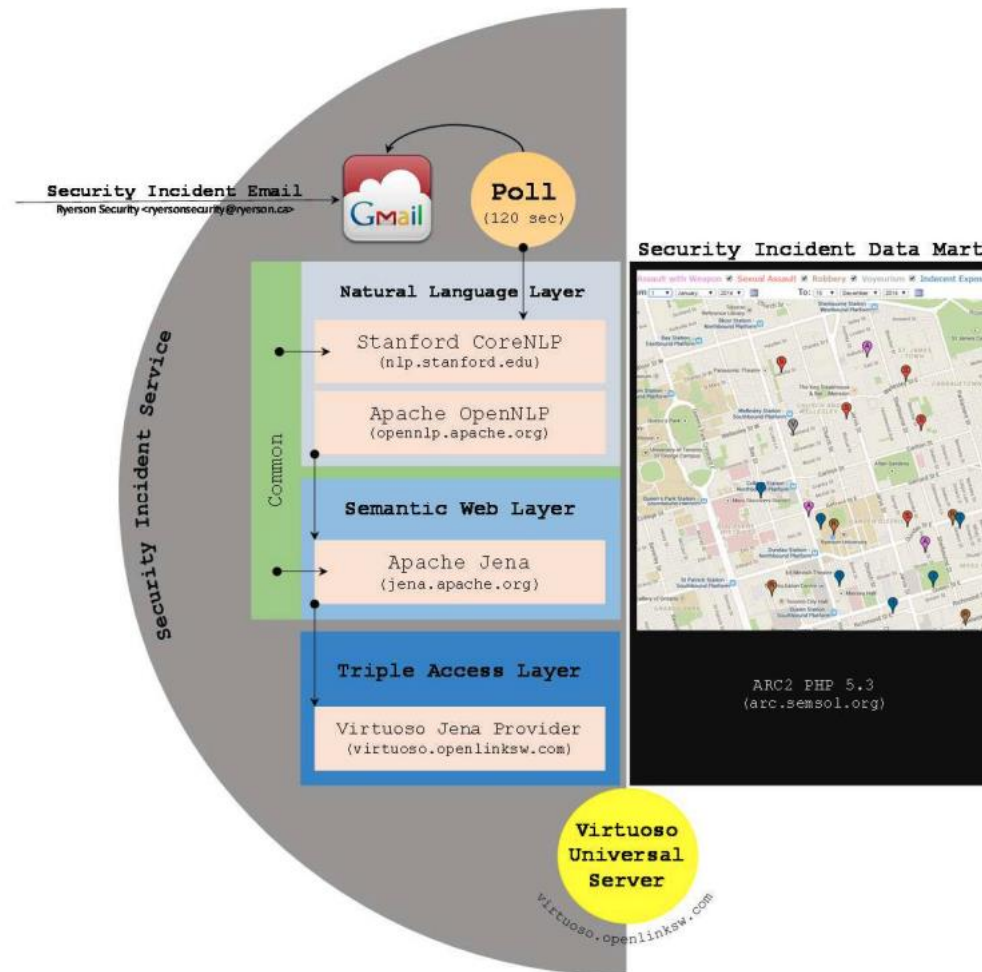


Figure 3. Machine Readable Security Incident Notification software system architecture. The left hemisphere is the software service which polls each 2 minutes a gmail account, seurityincidentsemanticweb@gmail.com, which is subscribed to be notified for any security incident by Ryerson University Integrated Risk Management. After fetching new incidents, the Natural Language Layer parse the text and extract information to an object of type `SecurityIncident`, defined in `Common`, and hand it to next layer. The Semantic Web Layer serialize the populated object to OWL individuals and pass them to Triple Access Layer to store them in the Virtuoso triple store. The right hemisphere is the primitive dashboard at <http://semionet.rnet.ryerson.ca/sio/> which shows the incidents geographical distribution filtered by the date of event.

10

Team Project: GitHub Repo

Branding: Pick a name for project, design a logo, ...

Structured: Directories, Layers, Output, ...

Contributions: Pull Requests, Commits, ...

license.txt

readme.md: Explanation, setup, samples, ...

Let's have a look at good and bad repos

https://github.com/hosseinfani/security_incident_onto

<https://github.com/fani-lab/OpeNTF>

<https://github.com/fani-lab/LADy>

<https://github.com/fani-lab/Osprey>



11

A Brightspace Tour

 bright.uvic.ca/d2l/home/513675

Fall 2026 CSC 501 A01 A02 X

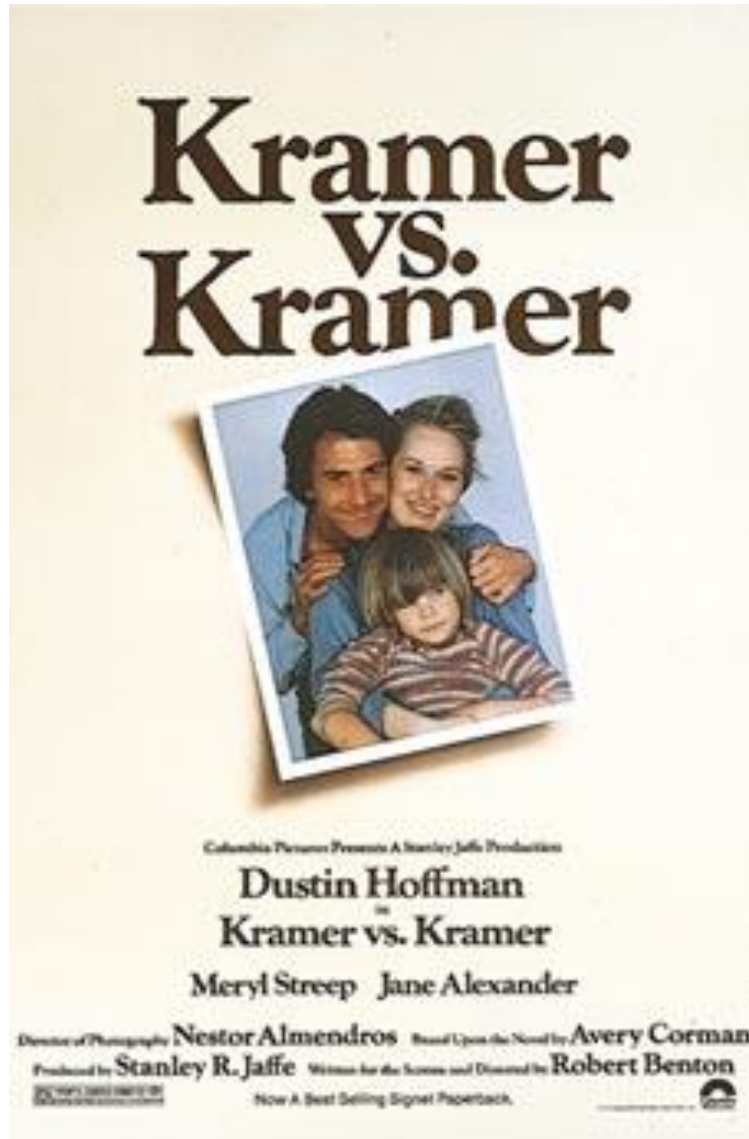
Hossein Fani
as Learner[Course Home](#) [Content](#) [Classlist](#) [Grades](#) [Class Progress](#) [Course Tools](#) [UVic Resources](#) [Help](#)

CSC501: Algorithm and Data Models

[Outline](#)[Instructor & Assistant](#)[Schedule \(Weekly, Timeline\)](#)[Lectures \(Forum, Slides, Recordings, Hyperlinks, Resources, ...\)](#)[Project \(Forum, Milestones, Manuals, ...\)](#) Milestone 1: Proposal: Problem Definition & Team Formation[Exams \(Forum, ...\)](#)

12

Googler vs. Googler



Googler: Who Builds Google
Googler: Who Works for Google
Googler: Who Uses Google



Larry Page and Sergey Brin in 2003



14

Another Brick In The Wall, 1979, Pink Floyd

<https://www.youtube.com/watch?v=YR5ApYxkU-U>

15

A Feedback

QUALITY

2.0**COMP2650**

Mar 20th, 2025

For Credit: **Yes** Attendance: **Not Mandatory** Grade: **Incomplete** Textbook: **Yes**

Fani inspired me to not be another brick in the **wall**. Besides that, he cannot teach.

DIFFICULTY

4.0

INSPIRATIONAL

Helpful  0  0



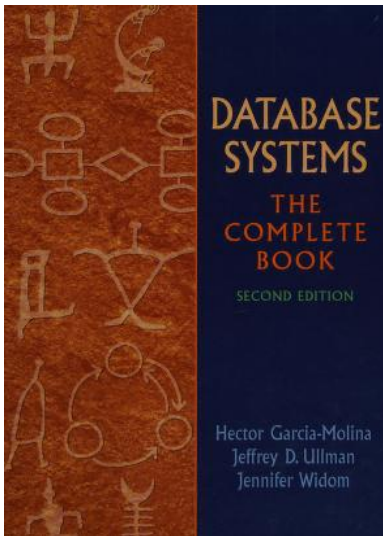
Take Aways

- Some great movies to watch ...
- Some wisdom on data modeling.
- Not to be another brick in the wall!



Conceptual Level | Entity-Relationship¹⁷Model

Outline: W02B: Thursday



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

III) Extended ER (EER)

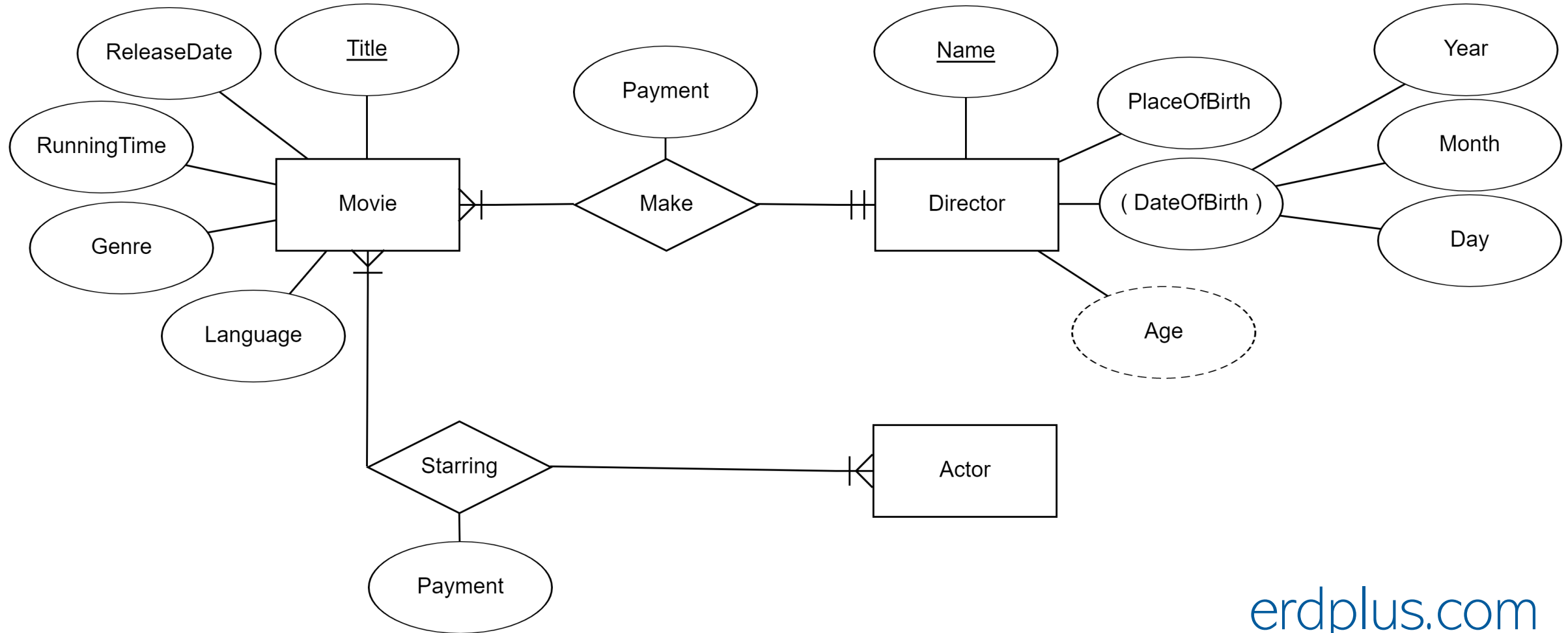
Entity-Relationship Model (ER)

1. Identify real-world **Entities, Attributes, Relationships**
2. Create graphical schema: ER Diagram (ERD)

A picture is worth a thousand words!

Standard → all data engineers all over the world understand ERD!

Entity-Relationship Model: Describe this domain ...



Entity-Relationship Model: Entity (e_i)

Real world thing that exists and is distinguishable from other things

e_1 =The Birds,

e_4 =Alfred Hitchcock,

e_7 =Tippi Hedren,

e_{10} =Kowwy,

e_{12} =Bruno Mars,

e_{14} =1 9Q4,

e_2 =Rosemary's Baby,

e_5 =Roman Polanski,

e_8 =Mia Farrow,

e_{11} =Hossein,

e_{13} =Taylor Swift,

e_{15} =Brothers Karamazov,

e_3 =L.A. Confidential

e_6 =James Cameron

e_9 =John Cassavetes

...

...

...

Entity-Relationship Model: Entity Set (E)

A set of entities of the same type that share the same properties

Movie	$=\{e_1:\text{The Birds}, e_2:\text{Rosemary's Baby}, e_3:\text{L.A. Confidential}, \dots\}$
Director	$=\{e_4:\text{Alfred Hitchcock}, e_5:\text{Roman Polanski}, e_6:\text{James Cameron}, \dots\}$
Actor	$=\{e_7:\text{Tippi Hedren}, e_8:\text{Mia Farrow}, e_9:\text{John Cassavetes}, \dots\}$
Instructor	$=\{e_{10}:\text{Kovvy}, e_{11}:\text{Hosseini}, \dots\}$
Singer	$=\{e_{12}:\text{Bruno Mars}, e_{13}:\text{Taylor Swift}, \dots\}$
Novel	$=\{e_{14}:\text{19Q4}, e_{15}:\text{Brothers Karamazov}, \dots\}$

Entity (~~Set~~)

23

Entity-Relationship Model: Entity Set (E)

A set of entities of the same type that share the same properties

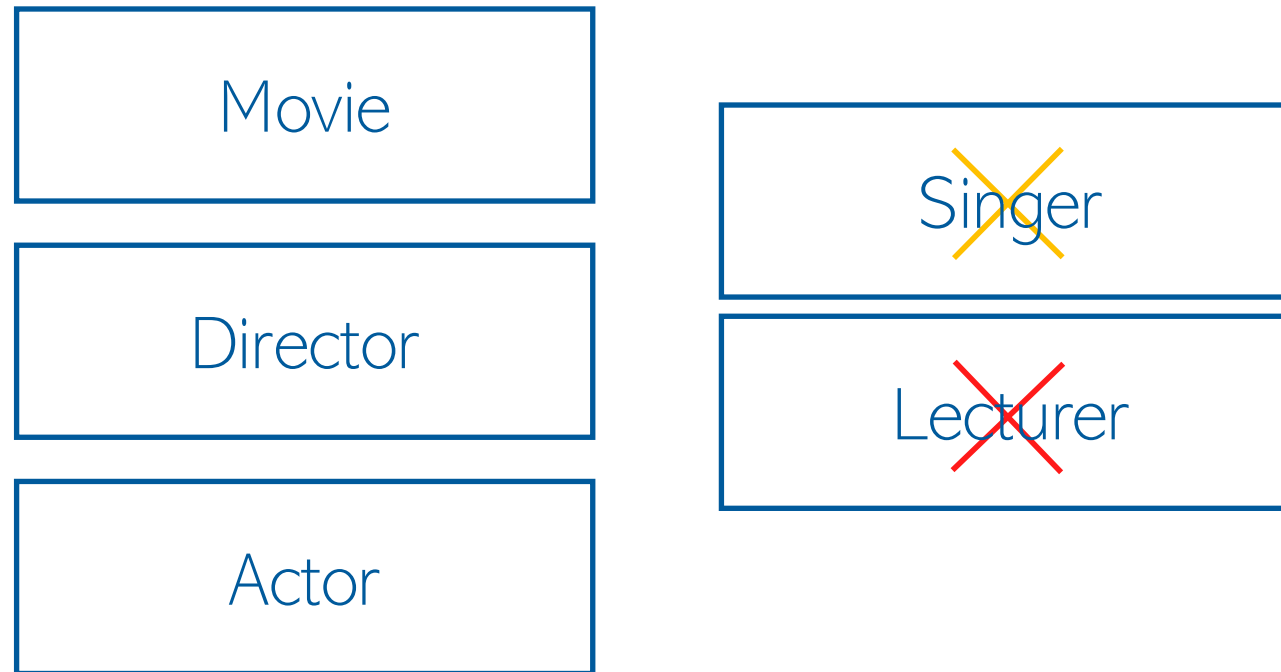
Movies

Directors

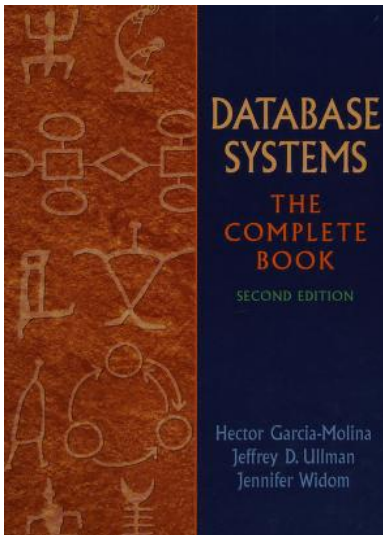
Actors

Entity-Relationship Model: Entity Set (E): Faithfulness

Design should be faithful to the specifications of the application



Outline: W02B



CH04 (2nd Ed.)

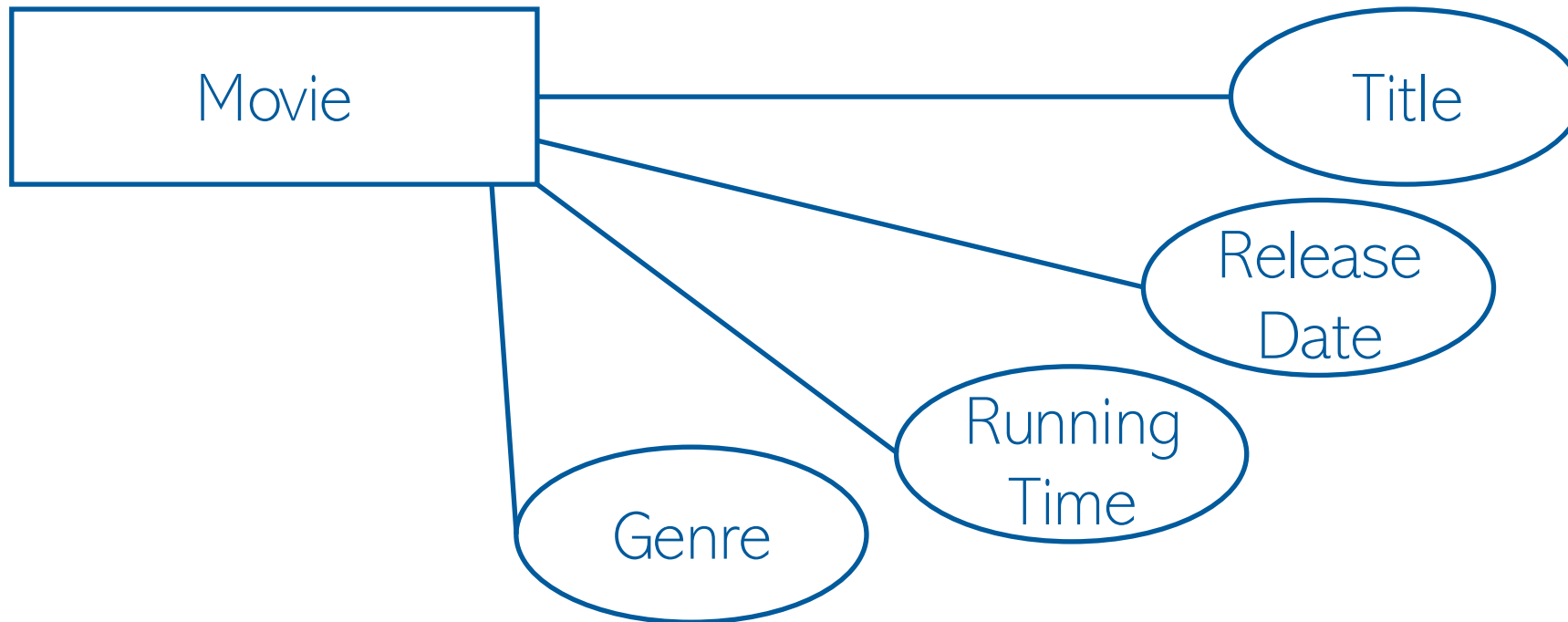
I) Entity

II) Attribute

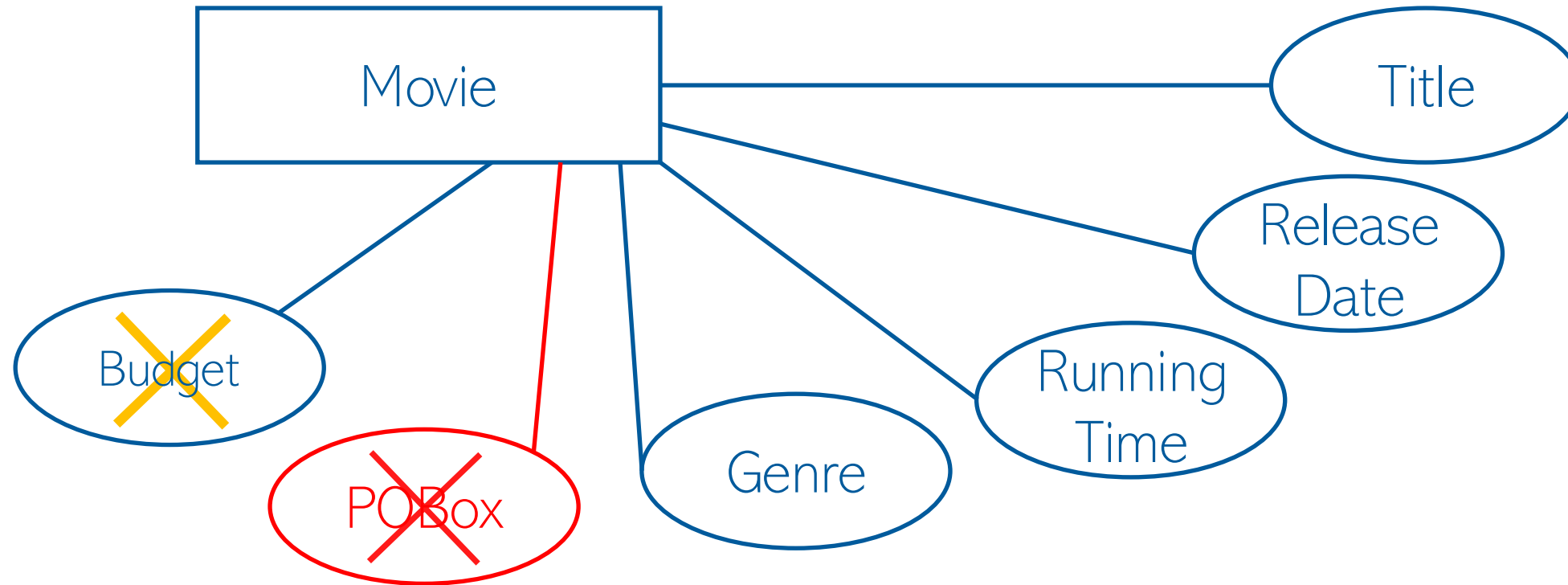
III) Relationship

IIII) Extended ER (EER)

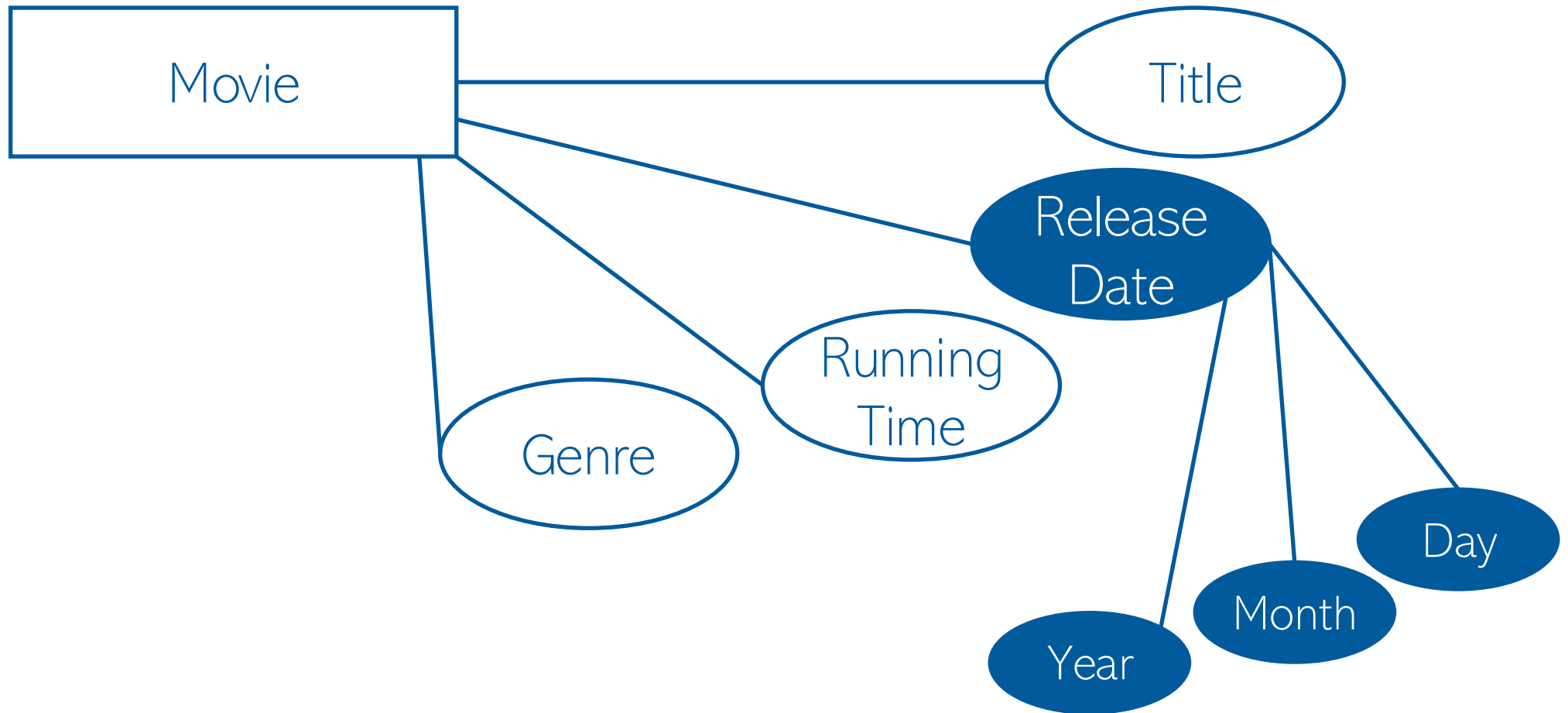
Attribute: Properties of Entities in Entity Set



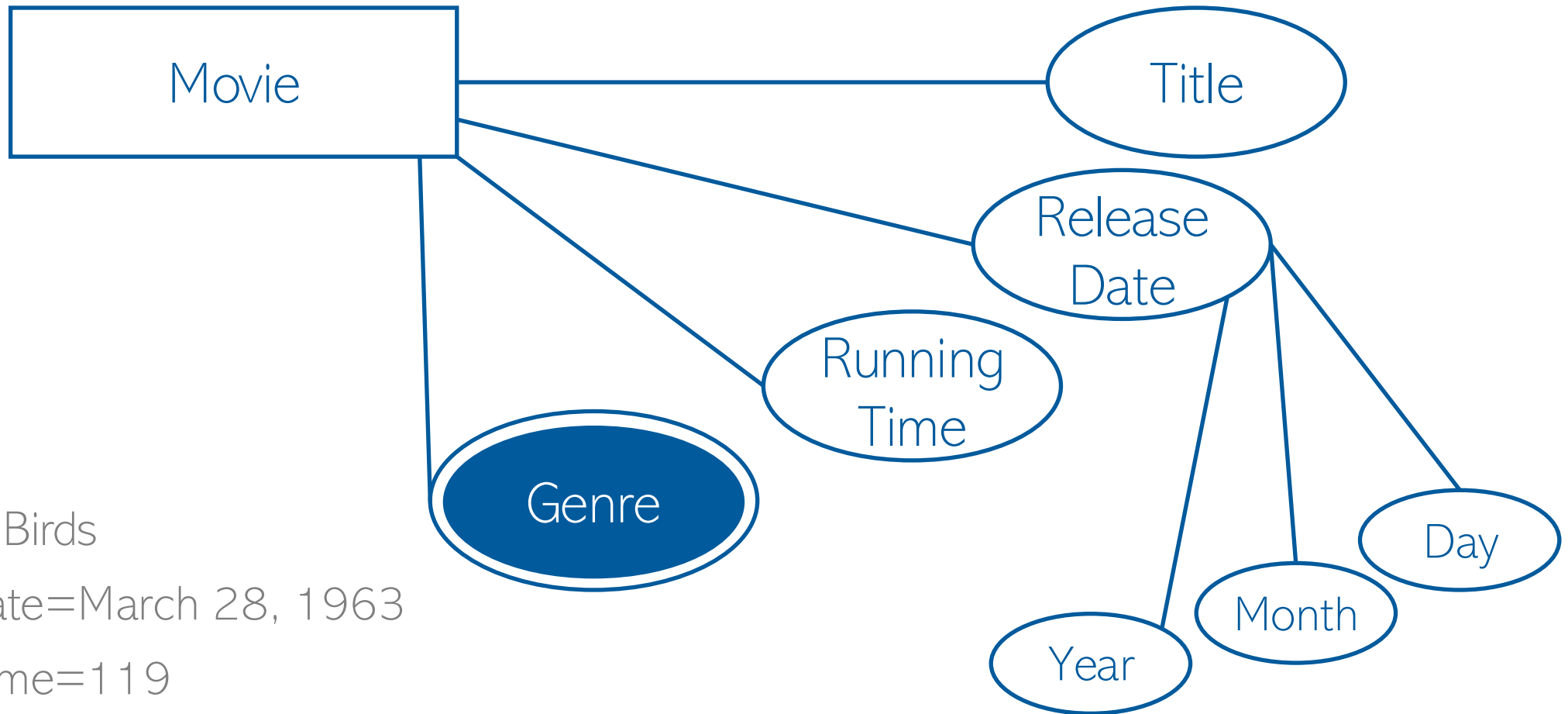
Attribute: Faithfulness



Attribute: Composite



Attribute: Multivalued



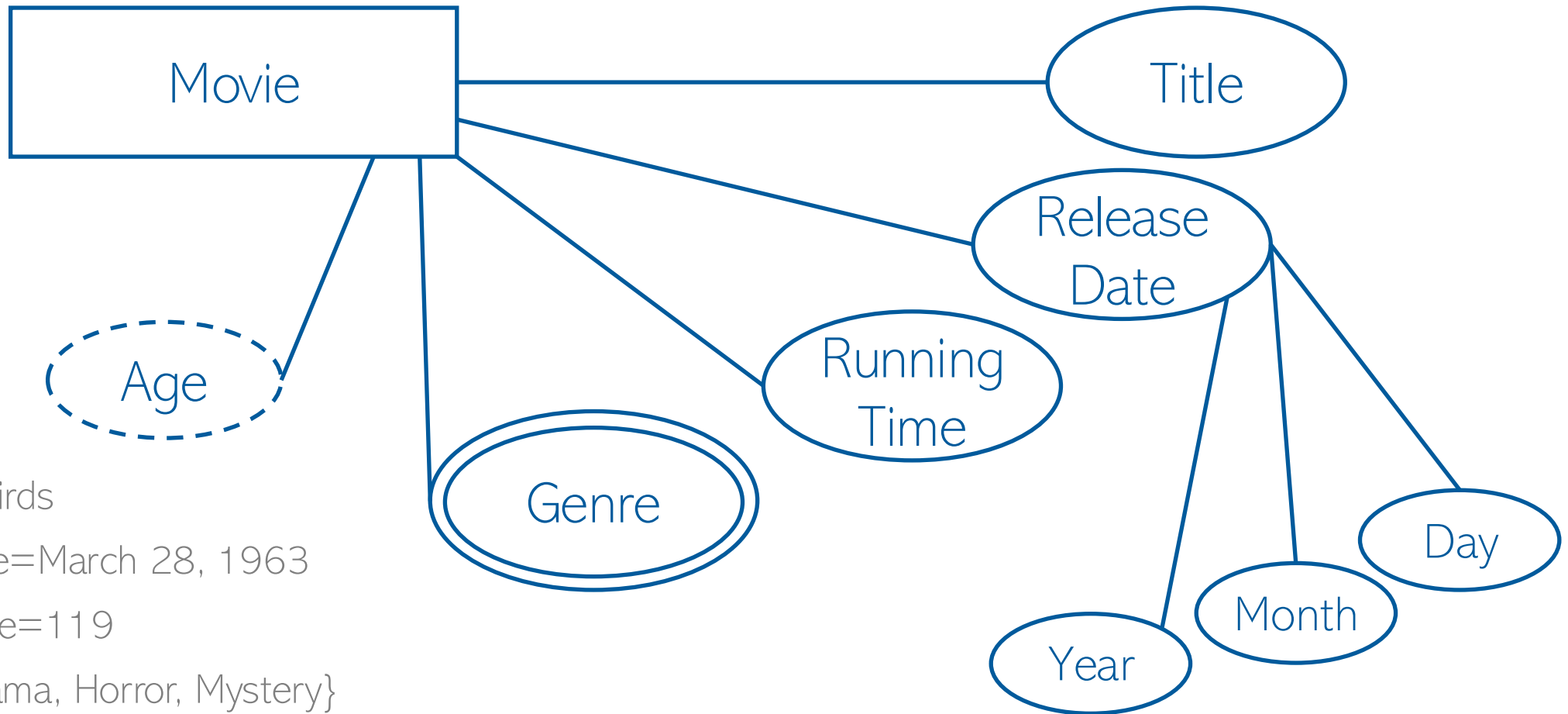
e_1 .Title=The Birds

e_1 .ReleaseDate=March 28, 1963

e_1 .RunningTime=119

e_1 .Genre={Drama, Horror, Mystery}

Attribute: Derived



$e_1.Title = \text{The Birds}$

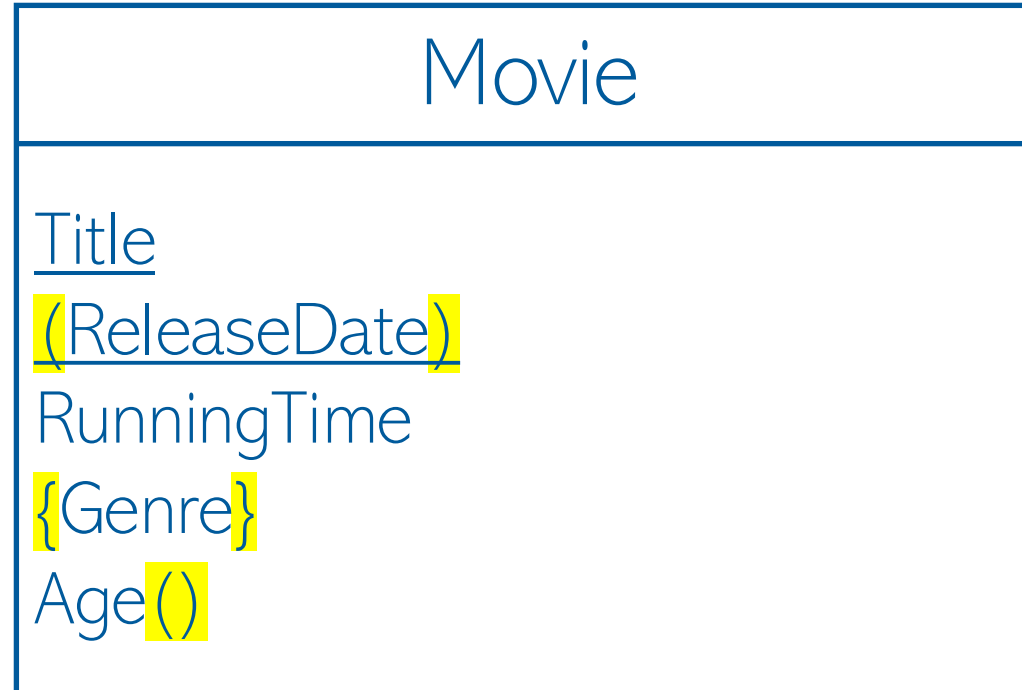
$e_1.ReleaseDate = \text{March 28, 1963}$

$e_1.RunningTime = 119$

$e_1.Genre = \{\text{Drama, Horror, Mystery}\}$

$e_1.Age = \text{YEAR}(e_1.ReleaseDate - \text{NOW})$

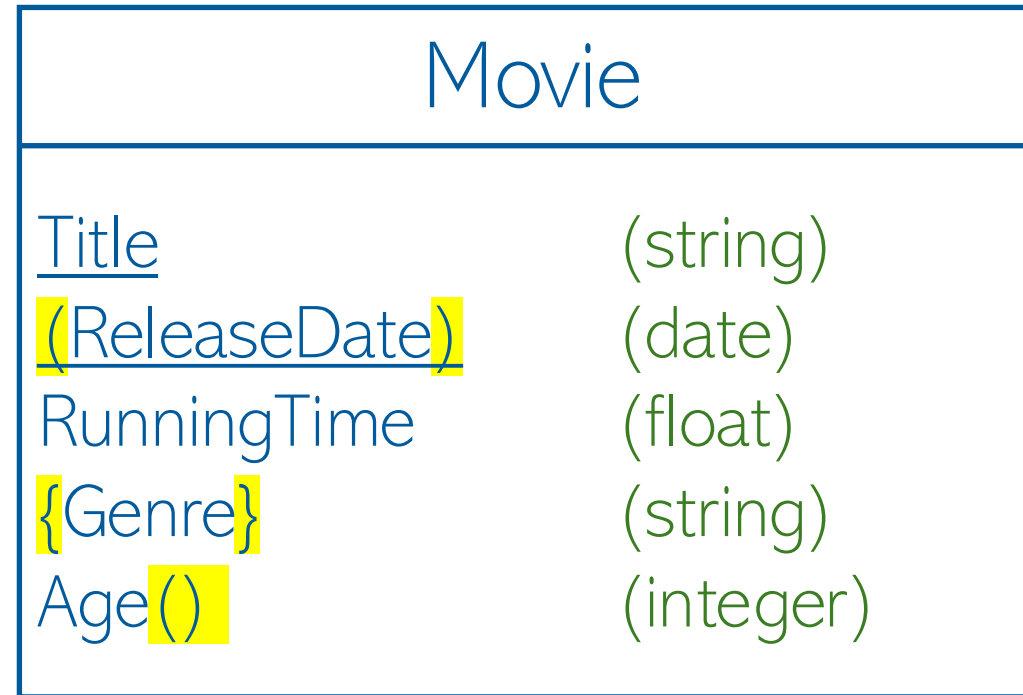
Entity + Attribute: Alternative Notation



32

Entity + Attribute: Alternative Notation: Data Type

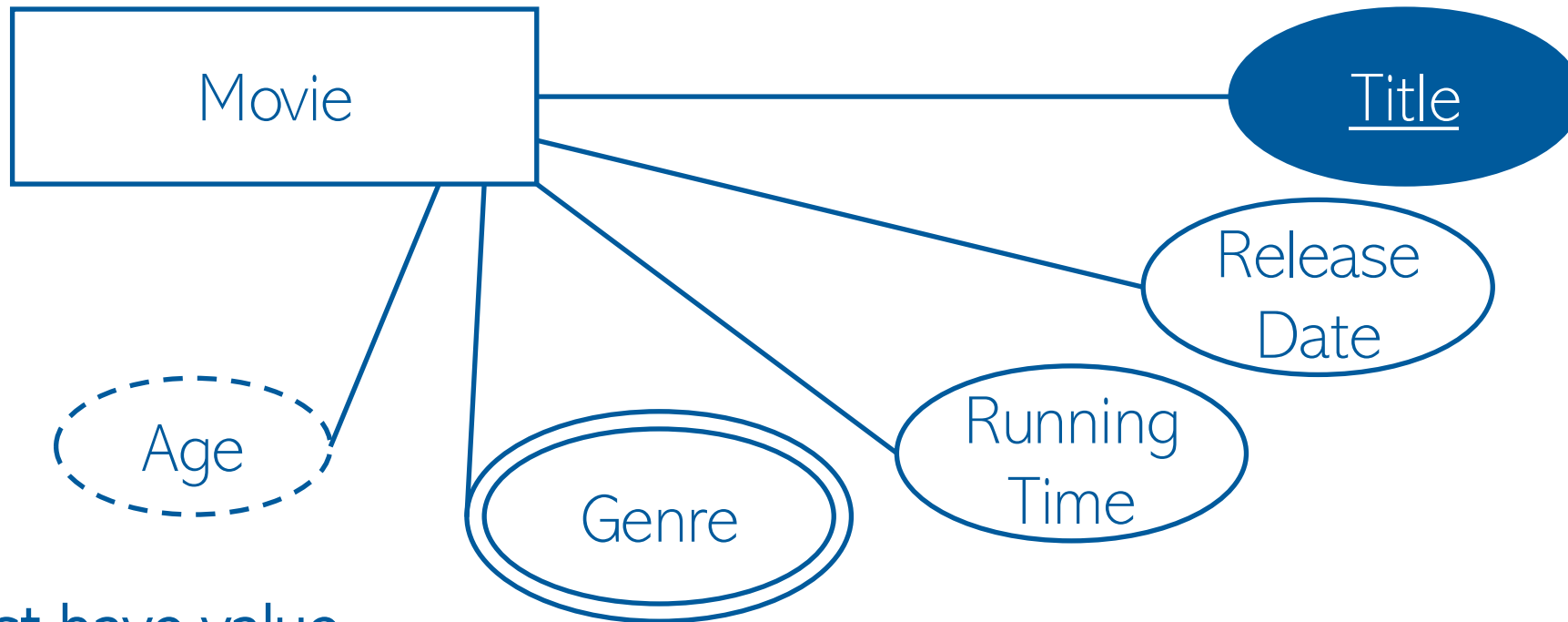
Standard E/R model does not have!



33

Attribute: **Key**

One or a set of attributes uniquely identify an entity in entity set



Key must have value.

A movie is identified by its Title.

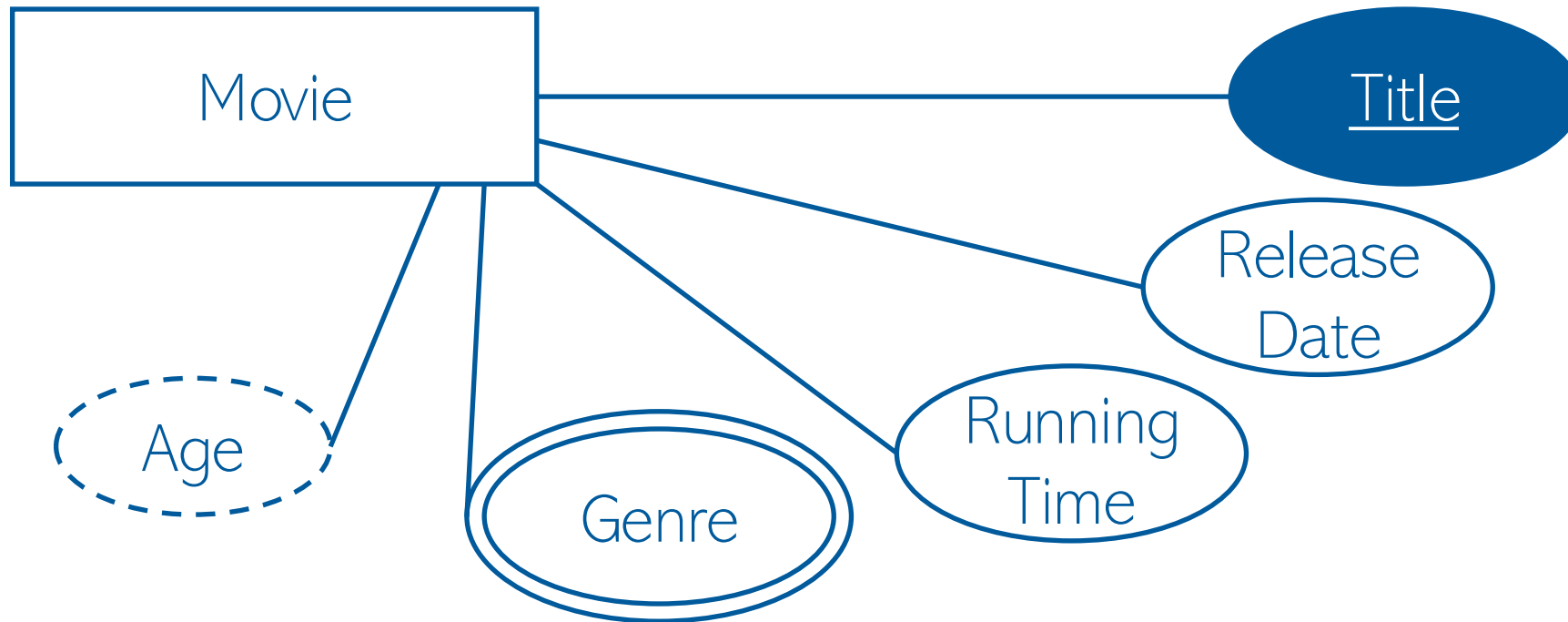
Two movies cannot have same value for Title.

The value in Title identifies one and only one movie!

34

Attribute: Key

One or a set of attributes uniquely identify an entity in entity set



Rosemary's Baby (1968) by Roman Polanski

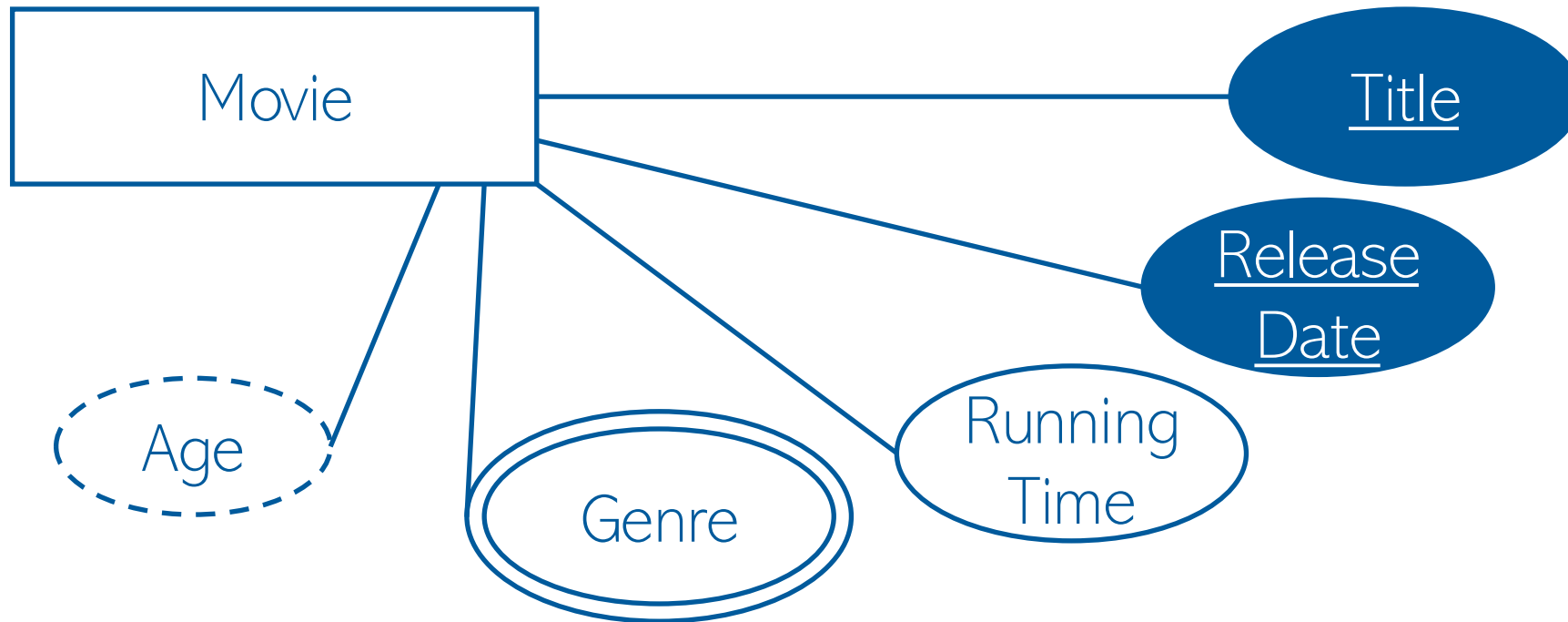
Rosemary's Baby (2014) by Agnieszka Holland (TV Mini-Series)

Rosemary's Baby (2015) by Dane Kissel

35

Attribute: Key

One or a set of attributes uniquely identify an entity in entity set



Rosemary's Baby (1968) by Roman Polanski

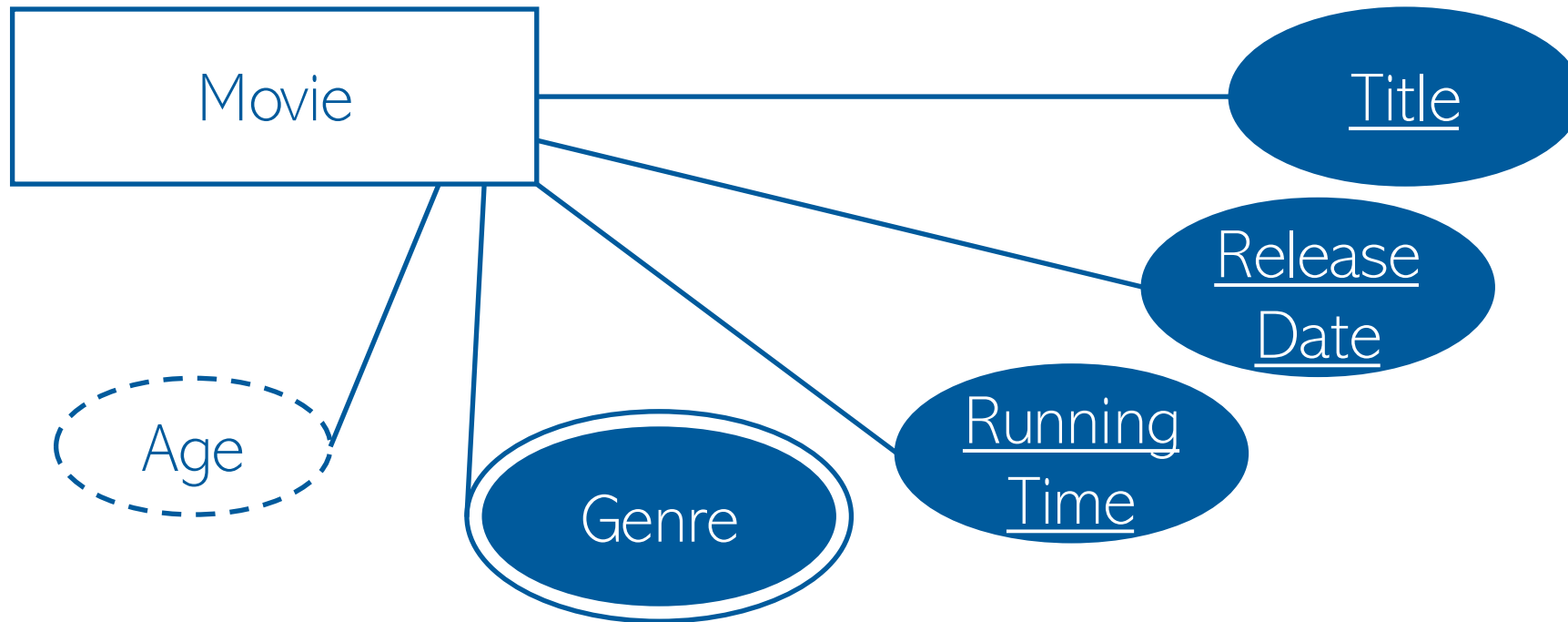
Rosemary's Baby (2014) by Agnieszka Holland (TV Mini-Series)

Rosemary's Baby (2015) by Dane Kessel

36

Attribute: Key

One or a set of attributes uniquely identify an entity in entity set



At worst case, all attributes together are the key.

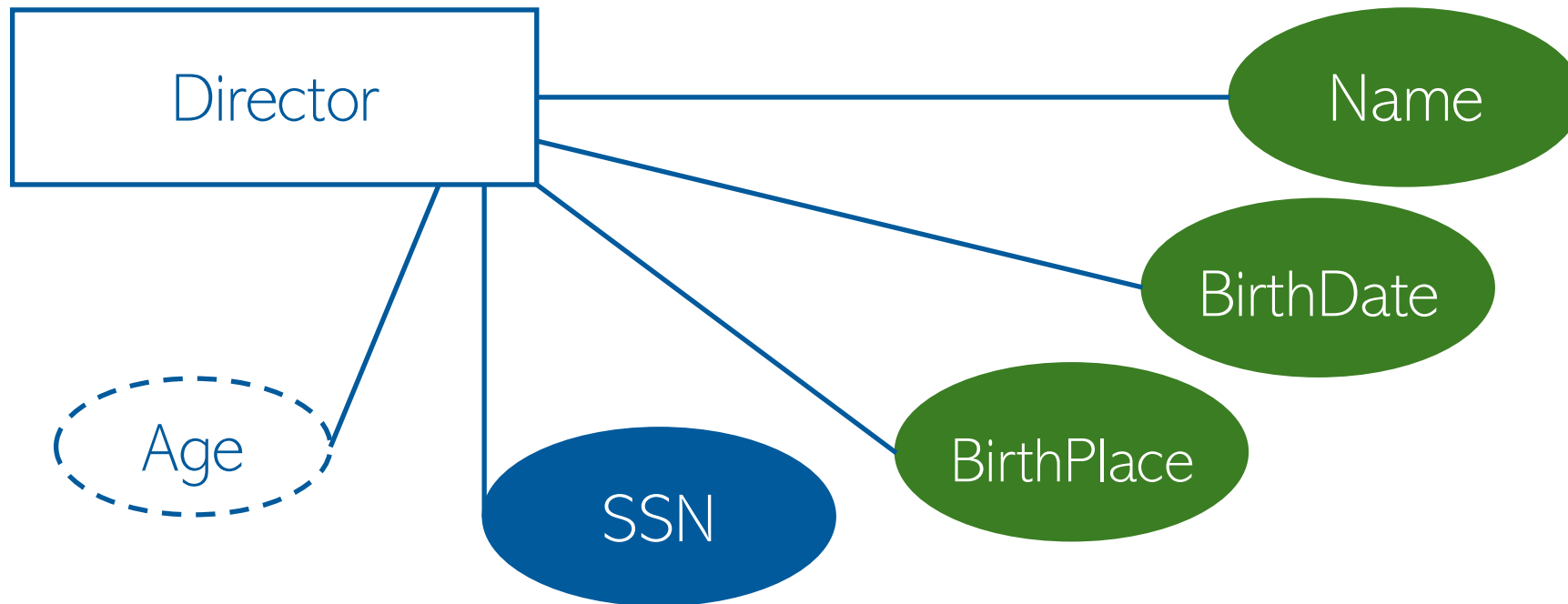
→ No duplicate entity in the entity set.

→ All attribute must have value.

37

Attribute: Multiple Keys

Standard E/R model does not have!



$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

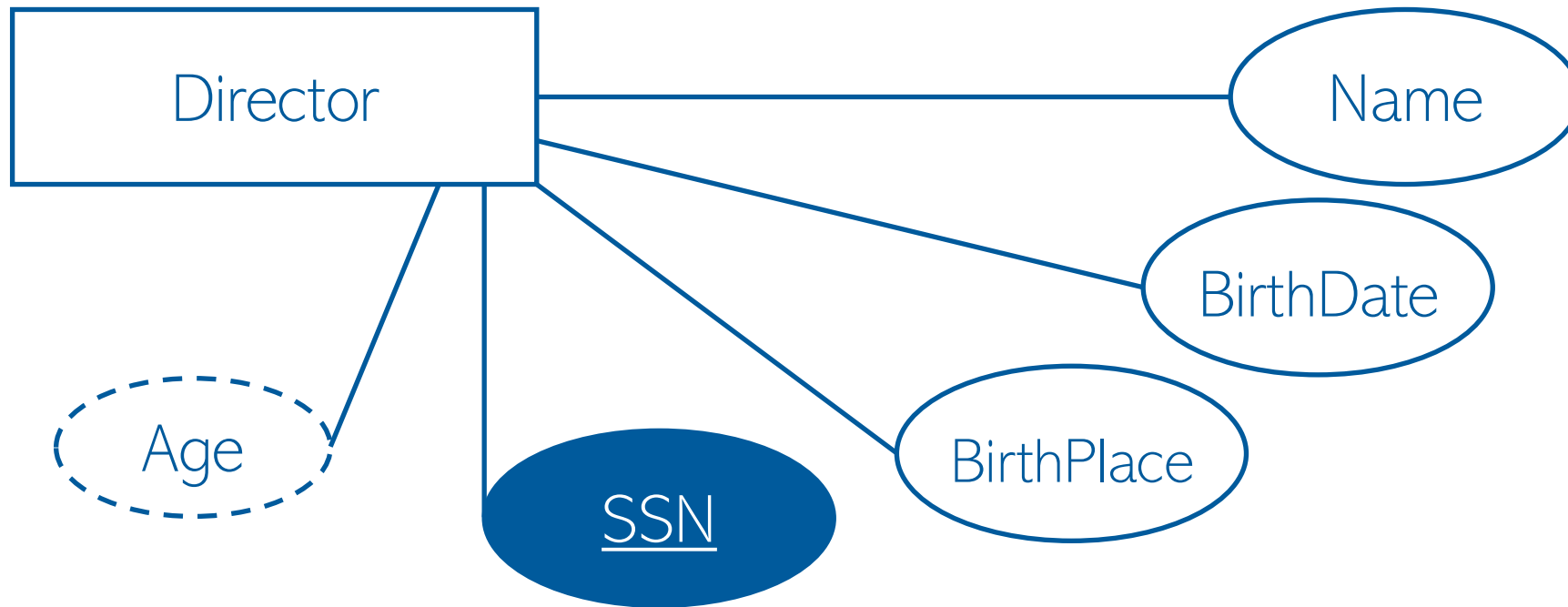
$K_2 = \{\text{SSN}\}$

Choose?

38

Attribute: Multiple Keys

Standard E/R model does not have!



$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

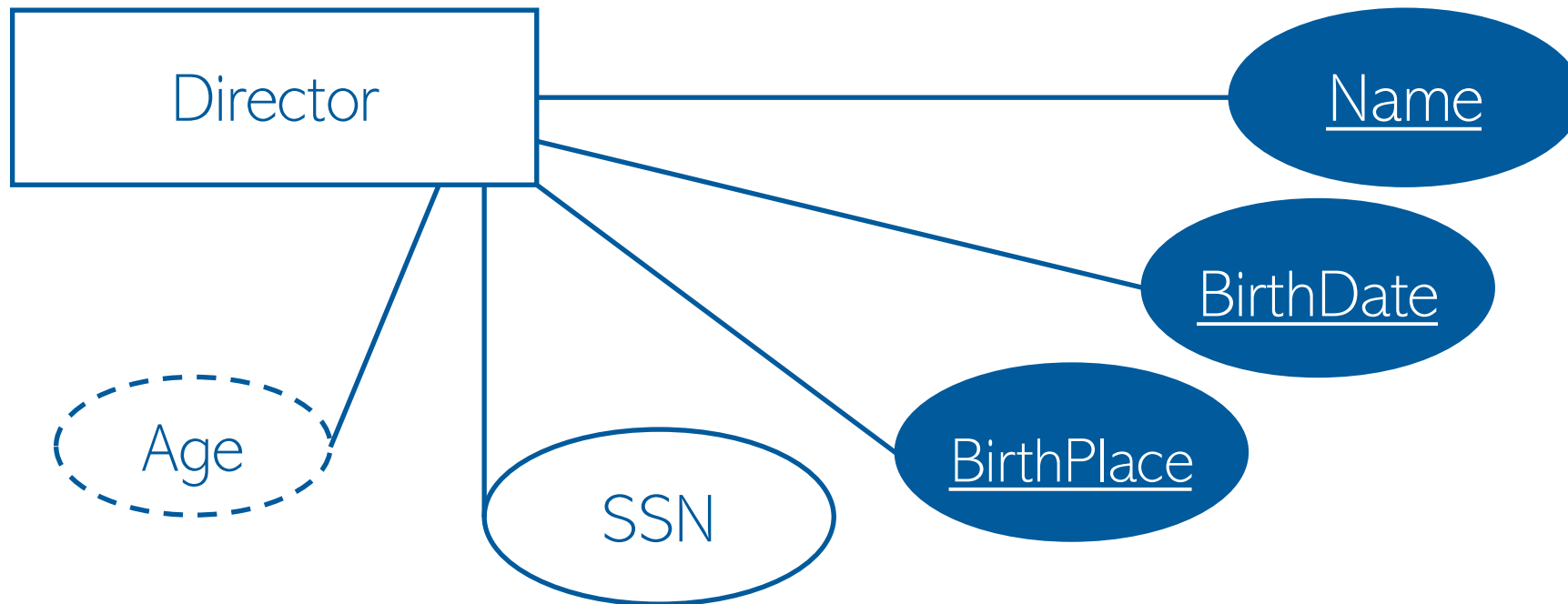
$K_2 = \{\text{SSN}\}$

Choose? Since simplicity matters, we may opt for K_2 as the Primary Key (PK)

39

Attribute: Multiple Keys

Standard E/R model does not have!

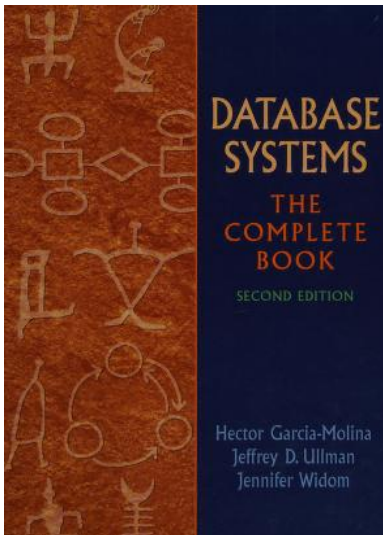


$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

$K_2 = \{\text{SSN}\}$

Choose? Since not all directors has SSN, we must select K_1 as the Primary Key

Outline: W02B



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

IIII) Extended ER (EER)

Relationship (r): Association among Entities of Different Sets

Movie (E_1)

e_1 .Title=The Birds

e_1 .ReleaseDate=March 28, 1963

e_1 .RunningTime=119

e_1 .Genre={Drama, Horror, Mystery}

e_1 .Age=51

Director (E_2)

e_2 .Name=Alfred Hitchcock

e_2 .BirthDate=August 13, 1899

e_2 .BirthPlace=England

e_2 .SSN=NULL

$r = (e_1, e_2) = (\{\text{The Birds, March-28-1963}\}, \{\text{Alfred Hitchcock, August-13-1899}\})$

Relationship (r): Association among Entities of Different Sets

Movie (E_1)

e'_1 .Title='Rosemary's Baby'

e'_1 .ReleaseDate=June 12, 1968

e'_1 .RunningTime=136

e'_1 .Genre={'Drama', 'Horror'}

e'_1 .Age=51

Director (E_2)

e'_2 .Name='Roman Polanski'

e'_2 .BirthDate=August 18, 1933

e'_2 .BirthPlace=France

e'_2 .SSN=NULL

$r' = (e'_1, e'_2) = (\{\text{Rosemary's Baby, June-12-1968}\}, \{\text{Roman Polanski, August-18-1933}\})$

Relationship Set (R)

$r = (e_1, e_2) = (\{\text{The Birds, March-28-1963}\}, \{\text{Alfred Hitchcock, August-13-1899}\})$

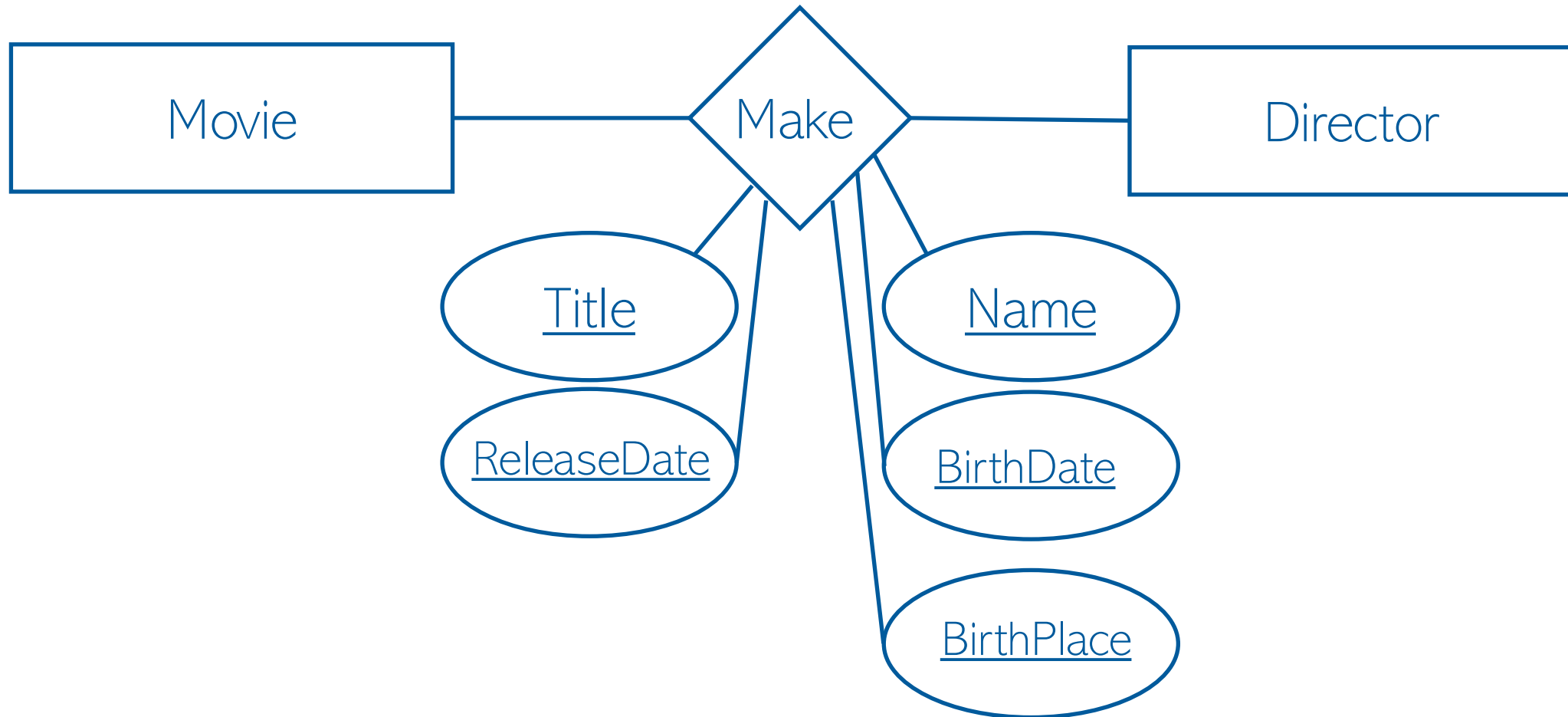
$r' = (e'_1, e'_2) = (\{\text{Rosemary's Baby, June-12-1968}\}, \{\text{Roman Polanski, August-18-1933}\})$

...

Binary Relationships

45

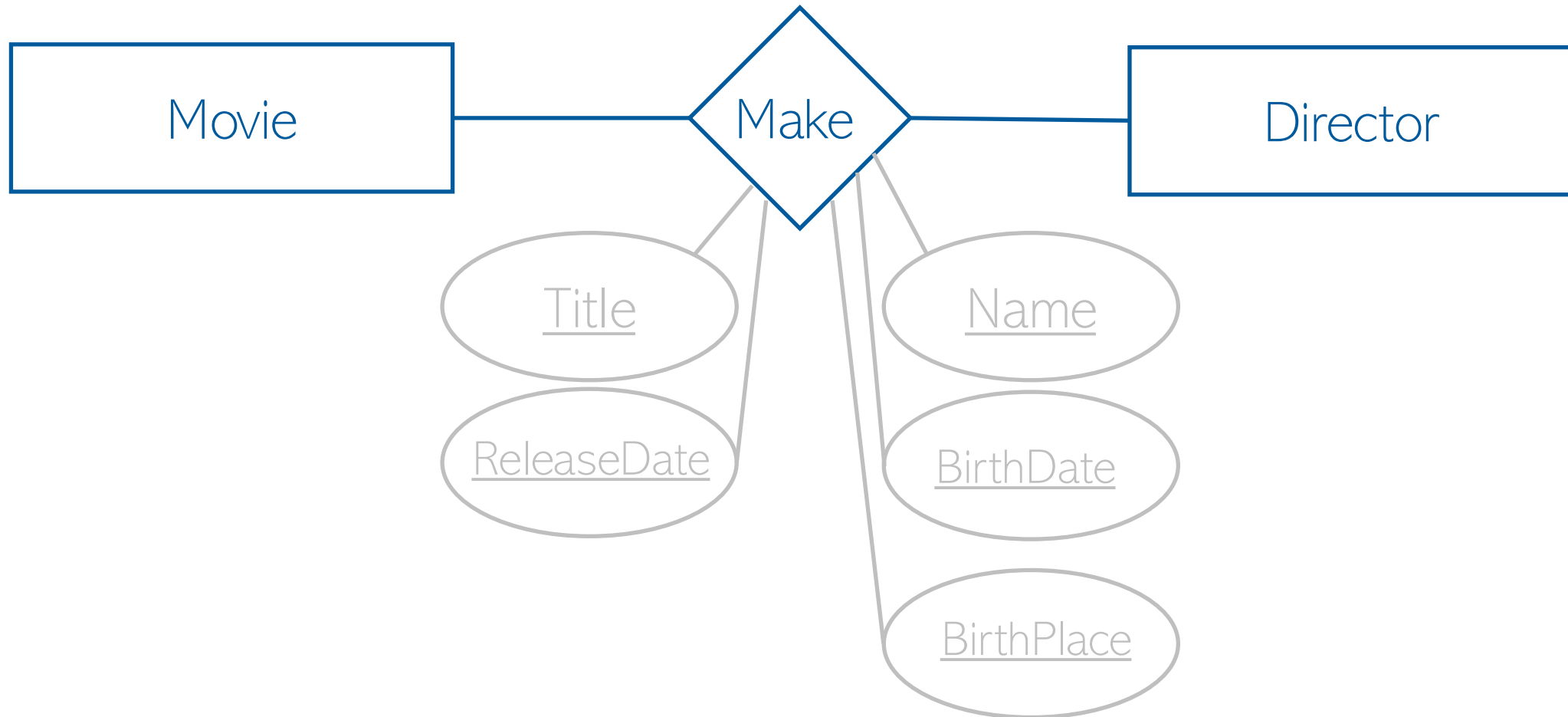
Relationship Set (R)



46

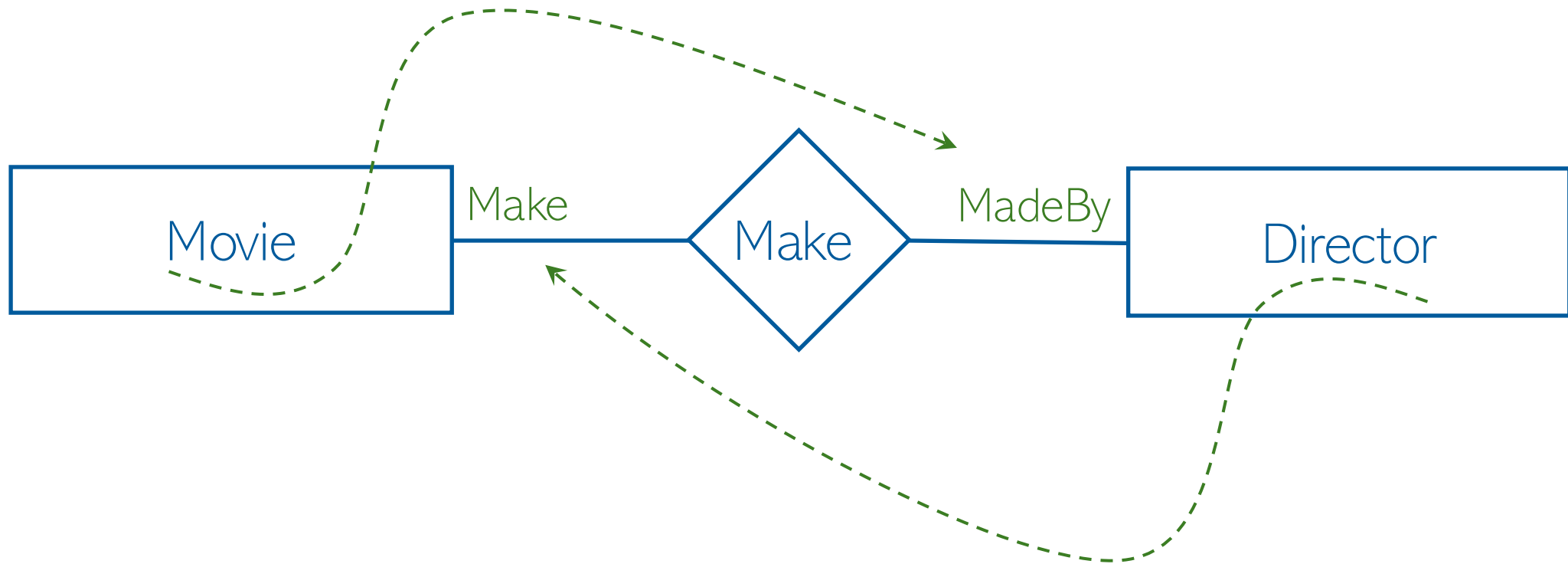
Relationship Set (R)

We do not show the keys from participating entity sets.



47

Relationship Set (R): Roles (when not clear)



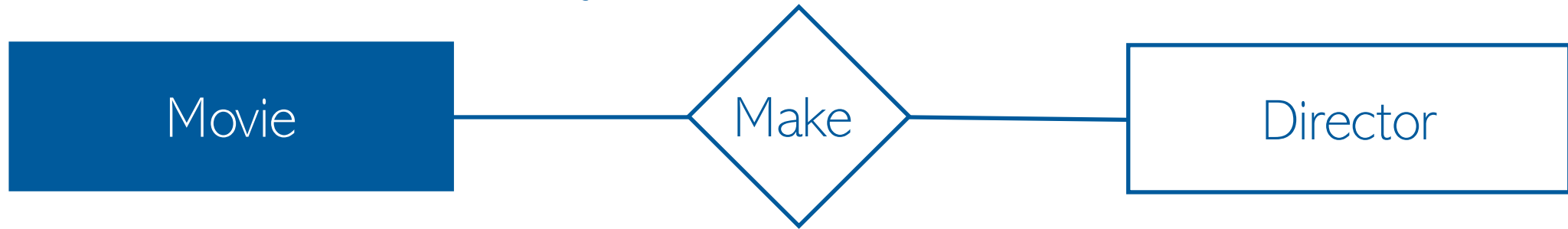
48

Relationship Set (R): Participation (Ordinality)

A Movie MUST participate in Make relationship?

A Movie MUST have Director?

Is there a Movie in the system without Director?



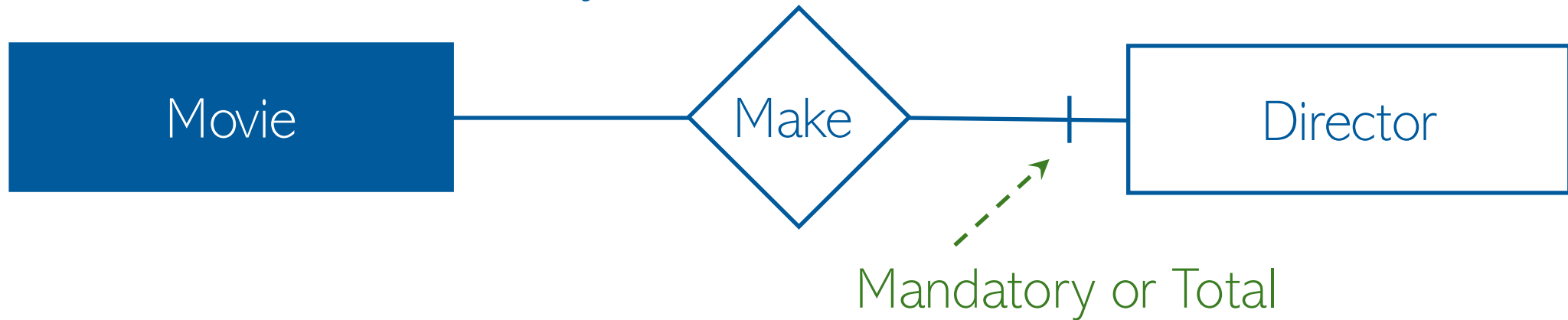
49

Relationship Set (R): Participation (Ordinality)

A Movie MUST participate in Make relationship? **Yes.**

A Movie MUST have Director? **Yes.**

Is there a Movie in the system without Director? **No!**



Movie's participation in Make relationship with Director is mandatory (total).

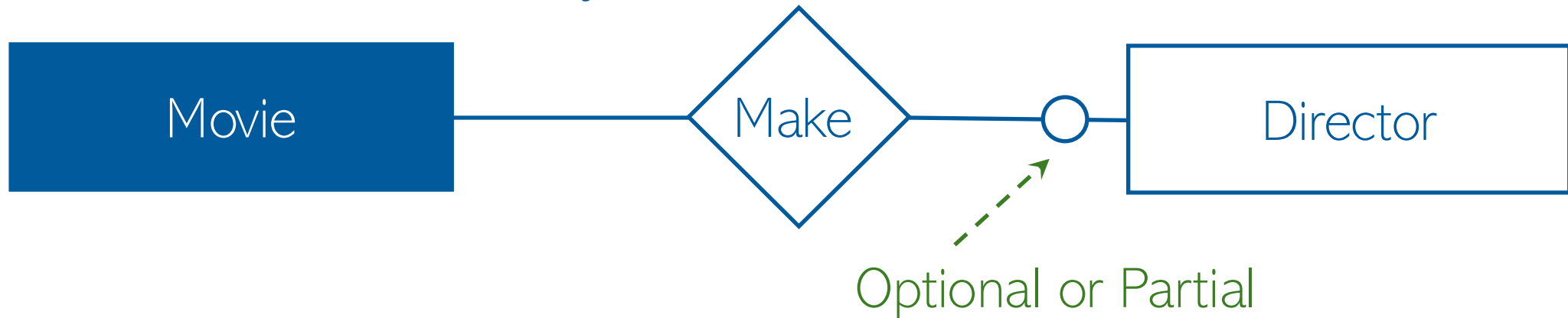
50

Relationship Set (R): Participation (Ordinality)

A Movie MUST participate in Make relationship? No!

A Movie MUST have Director? No!

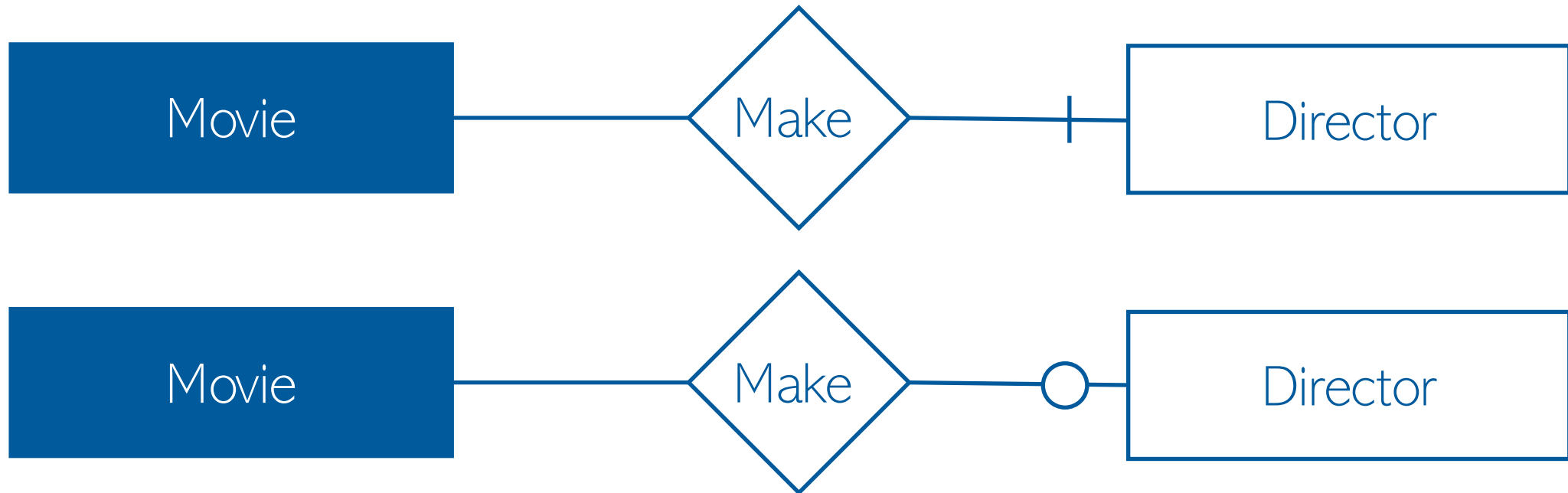
Is there a Movie in the system without Director? Yes.



Movie's participation in Make relationship with Director is optional (paartial).

51

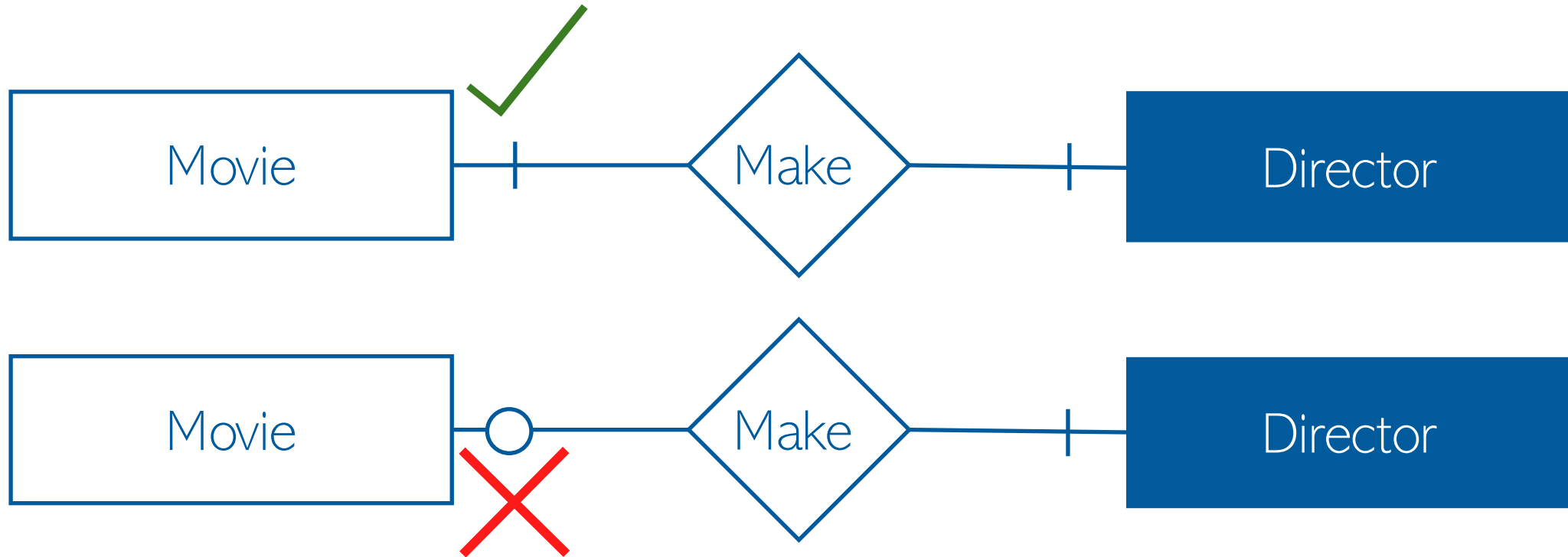
Relationship Set (R): Participation (Ordinality)



Design choice but needs justification ...

52

Relationship Set (R): Participation (Ordinality)



Relationship Set (R): Participation (Ordinality)

Minimum number of times an entity in an entity set can be associated with another entity in the related entity set.

Either	0: Optional, Partial
Or	1: Mandatory, Total

54

Relationship Set (R): Multiplicity (Cardinality)

Maximum number of times an entity in an entity set can be associated with another entity in the related entity set.

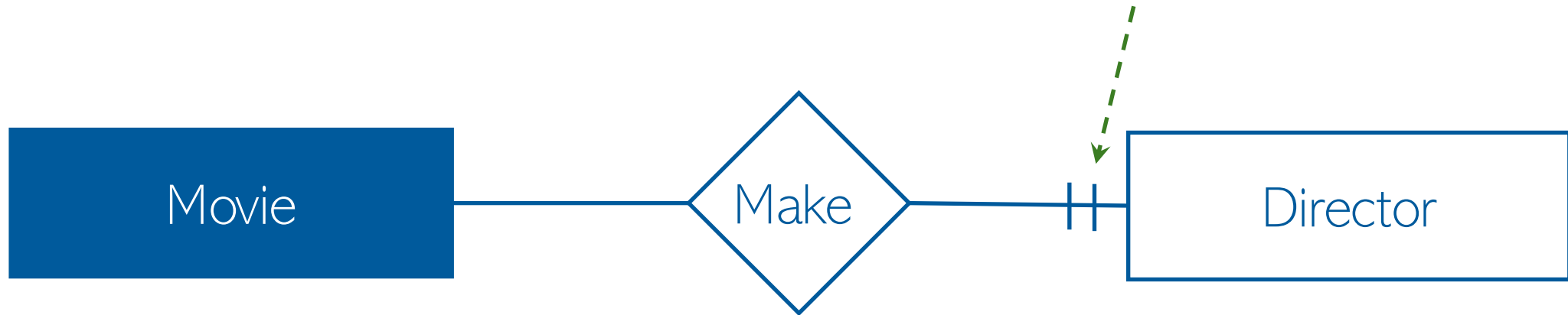
Either 1

Or Many (more than 1)

55

Relationship Set (R): Multiplicity (Cardinality)

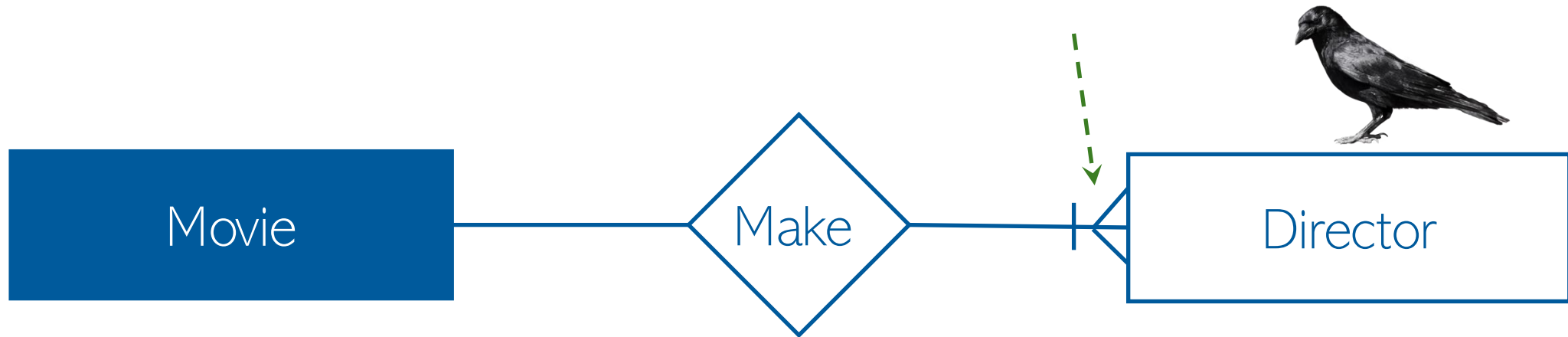
A Movie participate in Make relationship with 1 Director at max



56

Relationship Set (R): Multiplicity (Cardinality)

A Movie participate in Make relationship with many Director at max



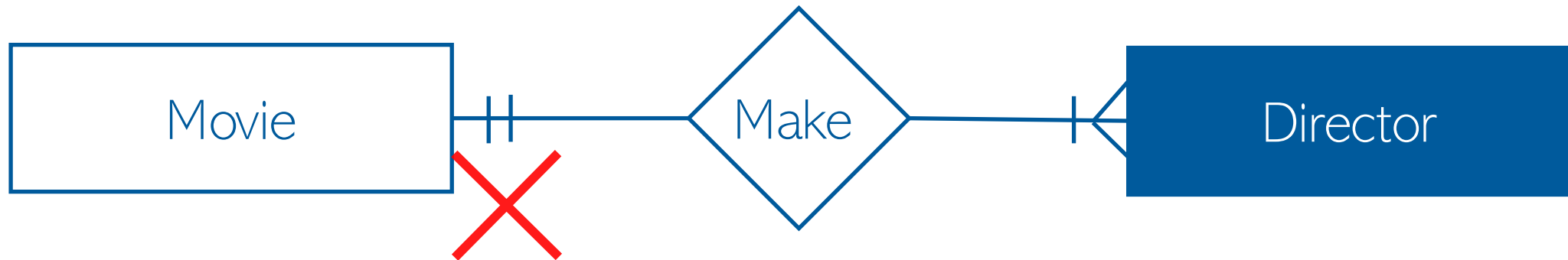
Barton Fink
Fargo
O Brother, Where Art Thou?
The Man Who Wasn't There
No Country for Old Men
Burn After Reading
A Serious Man
True Grit
Inside Llewyn Davis
Unbroken
Bridge of Spies
Hail, Caesar!



58

Relationship Set (R): Multiplicity (Cardinality)

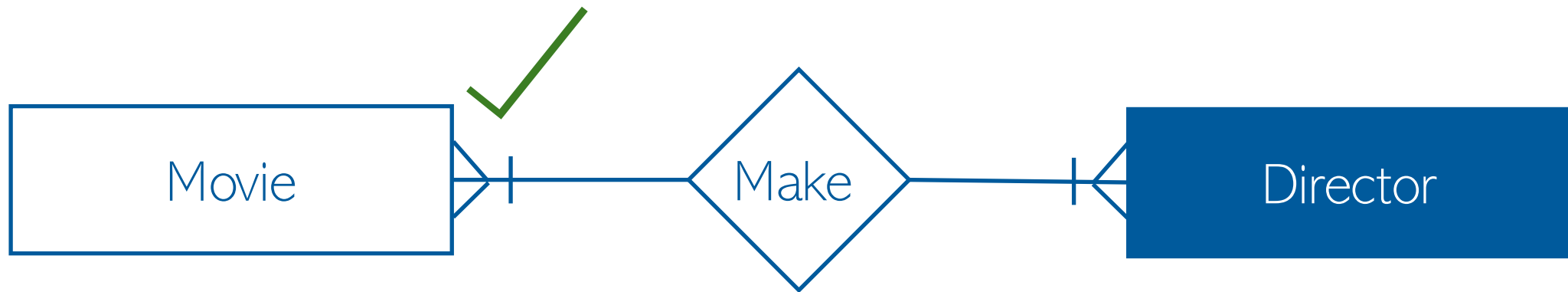
A Director participate in Make relationship with 1 Movie at max



59

Relationship Set (R): Multiplicity (Cardinality)

A Director participate in Make relationship with many Movie at max

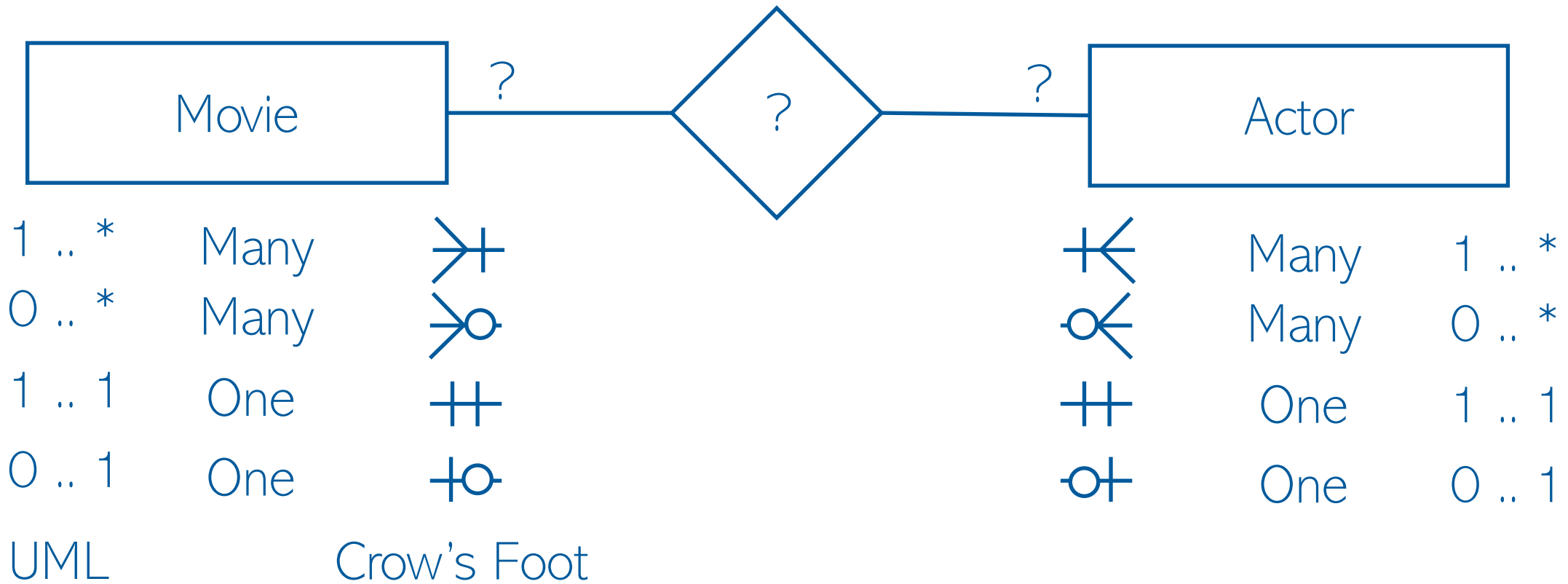


A Director entity can Make more than 1 Movie entity.

Many-Many Relationship (Set)

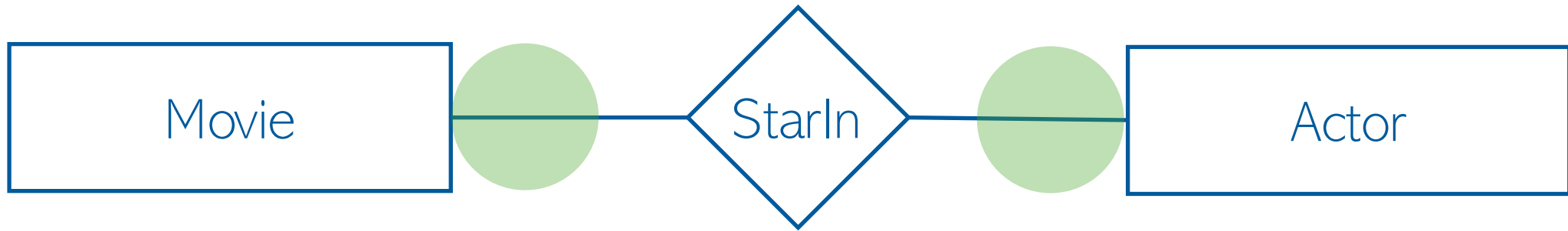
60

Relationship Set (R): Multiplicity (Cardinality)



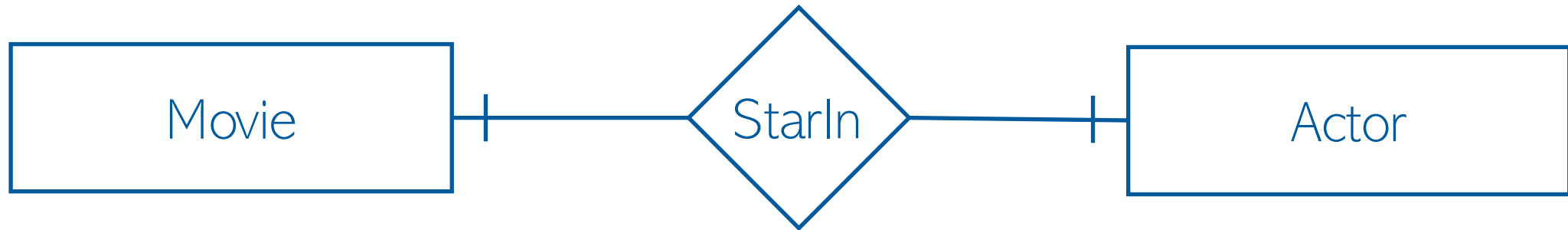
61

Relationship Set (R): Participation and Multiplicity



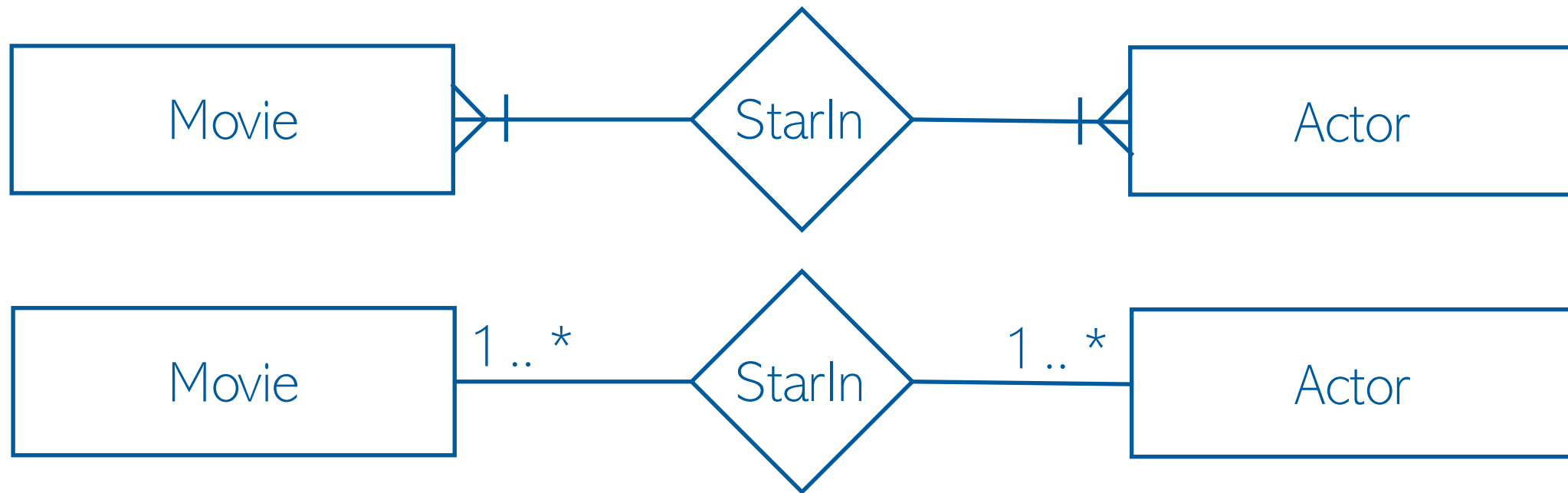
62

Relationship Set (R): Participation and Multiplicity



A Movie must have at least 1 Actor entity (Mandatory).
An Actor must have at least 1 Movie entity (Mandatory).

Relationship Set (R): Participation and Multiplicity



A Movie may have more than 1 Actor entity (Many).
An Actor may have more than 1 Movie entity (Many).

ALFRED HITCHCOCK S

